

Investigating Spontaneous Symmetry Breaking in IKKT Matrix Model

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in Science*



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Certificate of Examination

This is to certify that the dissertation titled **Investigating Spontaneous Symmetry Breaking in IKKT Matrix Model** submitted by **Piyush Kumar** (Reg. No. MS17051) for the partial fulfillment of BS-MS Dual Degree programme of the institute, has been examined by the thesis committee duly appointed by the institute. The committee finds the work done by the candidate satisfactory and recommends that the report be accepted.

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Declaration

The work presented in this dissertation has been carried out by me under the guidance of Dr. Anosh Joseph at the Indian Institute of Science Education and Research (IISER) Mohali.

This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bonafide record of original work done by me and all sources listed within have been detailed in the bibliography.

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In my capacity as the supervisor of the candidates project work, I certify that the above statements by the candidate are true to the best of my knowledge.

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Abstract

In this thesis, we studied the dimensionality of emergent spacetime via spontaneous breaking of $SO(10)$ rotational symmetry (SSB) in the IKKT matrix model, which is proposed as a constructive definition of Type IIB superstring theory, using complex Langevin method (CLM). First, we identified the challenges inherent to the simulations of matrix models using CLM and explored ways to overcome them. In particular, we use the deformed model to cure singular drift problem and make the extrapolations as the deformation parameters go to zero. Using the deformation technique, we successfully performed simulations based on CLM and studied SSB in both bosonic and supersymmetric IKKT matrix model using spacetime extents as the order parameters. In the bosonic model, we did not observe SSB, whereas in the supersymmetric model, we did observe signs of SSB in terms on the non-equivalence of spacetime extents. We also discuss another deformed model which can be used to investigate SSB in IKKT matrix model.

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Chapter 1

Introduction

Our modern understanding of the laws of nature rests on the two pillars of theoretical physics, namely, general relativity and quantum mechanics, developed in the twentieth-century. General relativity explains the large scale physics remarkably well, whereas quantum mechanics, as the name suggests describes physics at the microscopic level. Quantum field theory (a framework which combines special theory of relativity and quantum mechanics) has allowed a consistent unification of three fundamental forces, *viz.* the electromagnetic, the strong nuclear force and the weak nuclear force, culminating in a structure known as the ‘Standard Model’ of elementary particles. Experiments to date have shown excellent agreement with the predictions arising from these theories. The observations of gravitational waves from a black hole merger by LIGO in 2015 [Abbott 16] and the shadow of a supermassive black hole by Event Horizon Telescope (EHT) in 2019 [Akiyama 19] are a few of such instances for general relativity. On the other hand, the discovery of the Higgs boson at the LHC in 2012 [Aad 12] has fostered the validity of QFT as the theory of elementary particles. Although, quantum field theory and general relativity have proven to be the most successful theories in their regime, a consistent unification of the two is lacking. Much of the present research in theoretical physics is directed towards the reconciliation of gravity and quantum field theory is essential to address some of the most fundamental questions, including cosmological singularities and black hole information paradox, where quantum aspects of spacetime become important. The naive quantization general relativity within the framework of perturbative quantum field theory leads to a non-renormalizable theory.

String theory, which was originally proposed to understand the strong nuclear force, arose as a promising candidate for quantum theory of gravity, due to the presence of massless spin-2 particle, *graviton*, in the string spectrum. In other words, general relativity, is naturally incorporated in the theory at ordinary energies, with modifi-

cations arising only at very high energies/short distances. For string theory to be realistic, supersymmetry, that relates bosons to fermions and vice-versa, is essential, and the theory so obtained is known as Superstring theory. One of the salient features of superstring theories is the requirement of extra dimensions (six, to be precise, for the five classes of superstring theories namely, Type I, Type IIA/B, Heterotic $SO(32)$, $E_8 \times E_8$; and seven for M-theory) of space for its consistency, which raises the obvious question, 'why don't we see them?' A natural possibility for the existence of only four macroscopic dimensions, is that the six/seven extra dimensions are compactified, with size small enough to have escaped from experimental detection. This notion stems from the idea of Kaluza-Klein who tried to unify electromagnetism and general relativity by taking four macroscopic and one compact dimensions. The idea of 'curling up' of dimensions is convenient but the radius of compactification must be dynamically stabilized. Studying mechanisms of stabilization falls under the *moduli-space problem*. Moreover, within perturbative string theories, the space of $(3 + 1)$ -dimensional vacua, known as the *string landscape* is huge and thus non-perturbative aspects are important to identify the true vacuum of the theory. In a truly quantum theory of gravity, it is believed that the notion of spacetime becomes emergent, i.e., arising from non-gravitational degrees of freedom. Being used to the idea of taking a spacetime manifold to exist *a priori*, it is not immediately clear then what should be the basic degrees of freedom, but works from 1990s have proposed matrix degrees of freedom as a plausible option, to which we now turn.

Matrix models have become standard tools to study the non-perturbative aspects of superstring theories. The BFSS matrix model [Banks 97] is a quantum mechanical description of M-theory defined on eleven-dimensional spacetime in the infinite momentum frame. Soon afterward, the IKKT matrix model [Ishibashi 97] was proposed as a constructive formulation of Type IIB superstring theory in ten-dimensional spacetime. The BFSS and IKKT matrix model are obtained by performing the dimensional reduction of 10d $SU(N)$ super Yang-Mills theory to one and zero dimensions, respectively. The equivalence between matrix models and the superstring theory is obtained in the limit $N \rightarrow \infty$, where N is the size of the matrix. In this thesis, we will focus on the IKKT matrix model. Since, the IKKT model is defined on a point, it is the easiest matrix model for numerical studies using lattice regularization, because the continuum and lattice regularized versions of this theory are the same. In the context of IKKT model, the $N \times N$ matrices are thought of as the entities analogous to the coordinates on the spacetime, and the eigenvalues of the matrices as spacetime points. It is expected that smooth spacetime manifold emerges from the eigenvalues of the ten

bosonic matrices in the large N limit. Given the analogy, one can define observables using the eigenvalues of these matrices, to study the aspects of emergent spacetime. More concretely, we can investigate the distribution of eigenvalues of the matrices which naturally gives the extent of the corresponding dimension. The compactification of extra dimensions is then realized by the spontaneous symmetry breaking (SSB) of $\text{SO}(10)$ and $\text{SO}(9, 1)$ symmetry of the IKKT model in the Euclidean and Lorentzian signature, respectively.

Numerical studies of matrix models, or non-perturbative quantum field theories in general, involve Monte-Carlo (MC) methods, which work well as long as the action is real. MC is based on importance sampling method, which generates field configurations, $\{X\}$ from the configuration space, Ω , with a probability weight, $e^{-S[X]}$ (in Euclidean signature) and then the path integral is given by the statistical average over the sampled ensemble. Now, when the action becomes complex, the weight loses its probabilistic meaning since it is no longer positive definite and the importance sampling method fails. This is the notorious *sign-problem*. There are plenty of interesting systems with complex action like QCD with finite density and supersymmetric field theories to name a few. For IKKT model in the Euclidean signature, the Pfaffian of the fermionic matrix, obtained by integrating out the fermionic degrees of freedom is complex [Nishimura 00], causing the sign-problem hindering the studies using MC. A brute force method to circumvent the sign problem is *reweighting*, in which the imaginary part of the action is absorbed into the observable and the real part is used for generating field configurations based on importance sampling. There are two main shortcomings of reweighting procedure. First, the error and time required to estimate the physical quantity within a specific error limit increases exponentially with the size of the system; and second, the configurations with probability weight, $e^{-S[X]_R}$ is *irrelevant* to the system as a whole, unless the imaginary part of the action is small. To sum up, the sign-problem is NP-hard. Therefore, we need to resort to other methods for systems with complex action. Complex Langevin method (CLM) [Parisi 83, Klauder 84] and Lefschetz thimble method [Cristoforetti 12, Cristoforetti 13, Fujii 13] have shown promising results in evading the sign-problem. Both methods work by complexifying the dynamical field variables.

Previous numerical studies of the IKKT matrix model in Euclidean signature using Gaussian expansion method [Nishimura 11] and recently, using complex Langevin method, [Anagnostopoulos 20] have shown the SSB of $\text{SO}(10) \rightarrow \text{SO}(3)$. The presence of SSB has been attributed to the fluctuating phase of the Pfaffian since no SSB was

observed in the phase quenched models (or bosonic IKKT model, where we take the Pfaffian to be 1 by hand) [Hotta 99a]. The Lorentzian IKKT matrix model has also been studied numerically, implicating the emergence of expanding $(3 + 1)$ -dimensional universe, with exponential behaviour at early times and power-law behaviour at later times [Kim 12a, Kim 12b, Anagnostopoulos 12, Aoki 19, Hatakeyama 20].

1.1 Outline of the Thesis

In Chapter (2), we introduce the matrix regularization of membrane theory, *a la* Goldstone and Hoppe. In Chapter (3), we discuss the IKKT matrix model and the symmetries possessed by it. The relation with the perturbative Type IIB superstring theory is also made transparent. We also discuss the Wick rotation to obtain the Euclidean IKKT model. In Chapter (4), we discuss the foundations of the complex Langevin method and some of the technicalities associated with it, which must be taken into account to ascertain the reliability the the results so obtained. In Chapter (5), we discuss the essential ingredients for the application of CLM to IKKT matrix model and the observables to investigate the SSB of $SO(10)$ rotational symmetry. Chapters (6) and (7) are devoted to results, summary and future directions.

Chapter 2

Goldstone-Hoppe Regularization

In this chapter, we review the details of the Goldstone-Hoppe regularization procedure [Hoppe 89], which allows the mapping of continuous functions on membrane surface to finite sized matrices. More concretely, we have the following result: the Lie algebra of area preserving diffeomorphism (APD) on the surface, Σ_2 , i.e., $SDiff(\Sigma_2)$, is isomorphic to $\mathfrak{su}(n)$ in the limit $n \rightarrow \infty$. Originally, this correspondence was shown for 2-sphere but was later proven for surfaces of arbitrary topology.

To set up the terminology, let us start with what is meant by a membrane surface. We define *membranes* as a general M -dimensional surface sweeping out the $M + 1$ -dimensional volume in the D -dimensional spacetime, known as *target space*. For $M = 1(2)$, these objects are naturally called strings (membranes) and their action is taken to be proportional to worldsheet (worldvolume).

For the rest of this discussion, following Hoppe, we will take $M = 2$ and a particular surface, *viz.* $\Sigma = S^2$. As a first step, we look at the APD on a general surface by considering the following infinitesimal transformation of the surface coordinates $(\sigma_1, \sigma_2) \rightarrow (\tilde{\sigma}_1, \tilde{\sigma}_2)$

$$\tilde{\sigma}_1 = \sigma_1 + u_1(\sigma_1, \sigma_2), \tag{2.1}$$

$$\tilde{\sigma}_2 = \sigma_2 + u_2(\sigma_1, \sigma_2). \tag{2.2}$$

In order for this diffeomorphism to be APD, ($\det(J) = 1$, where J is the Jacobian matrix) it must preserve the Poisson Bracket¹, i.e.,

$$\{\tilde{\sigma}_1, \tilde{\sigma}_2\} = \{\sigma_1, \sigma_2\}, \tag{2.3}$$

¹Recall that in classical field theory, where canonical transformations preserve PB, or equivalently, the volume of phase space.

which is equivalent to

$$\frac{\partial u_1}{\partial \sigma_2} + \frac{\partial u_2}{\partial \sigma_1} = 0. \quad (2.4)$$

Now, consider a function $f(\sigma_1, \sigma_2)$ on the surface, and $(u_1, u_2) = (\partial f / \partial \sigma_2, -\partial f / \partial \sigma_1)$, which solves Eq. (2.4) identically. This gives us the following expression for the generators of APD

$$L_f = \left(\frac{\partial f}{\partial \sigma_2} \right) \frac{\partial}{\partial \sigma_1} - \left(\frac{\partial f}{\partial \sigma_1} \right) \frac{\partial}{\partial \sigma_2}. \quad (2.5)$$

It is trivial to check that $L_f \sigma_1 = u_1$ and $L_f \sigma_2 = u_2$. These generators satisfy the Lie algebra

$$[L_f, L_g] = L_{\{f, g\}}; \quad \text{with} \quad \{f, g\} = \frac{\partial f}{\partial \sigma_1} \frac{\partial g}{\partial \sigma_2} - \frac{\partial f}{\partial \sigma_2} \frac{\partial g}{\partial \sigma_1}. \quad (2.6)$$

Up to this point, the discussion holds for a general surface. Now moving on to S^2 , we can make the substitution, $\sigma_1 = \phi, \sigma_2 = \cos \theta$ and any function on S^2 can be expanded in the terms of spherical harmonics, $Y_{lm}(\sigma_1, \sigma_2)$, which means we can form a basis of the generators of APD as follows

$$L_A = \left(\frac{\partial Y_A}{\partial \sigma_2} \right) \frac{\partial}{\partial \sigma_1} - \left(\frac{\partial Y_A}{\partial \sigma_1} \right) \frac{\partial}{\partial \sigma_2}, \quad (2.7)$$

where we have used the notation A to represent the indices of spherical harmonics, (lm) . The basis of generators satisfy the following algebra

$$[L_A, L_B] = L_{\{Y_A, Y_B\}} = f_{ABC} L_C, \quad (2.8)$$

using $\{Y_A, Y_B\} = f_{ABC} Y_C$, where f_{ABC} are the structure constants defined by the Poisson brackets of the spherical harmonics, and $(A, B, C) = (lm, l'm', l''m'')$. Consider now the parameterization of S^2 in terms of Euclidean coordinates, $\{(\xi_1, \xi_2, \xi_3) : \xi_1^2 + \xi_2^2 + \xi_3^2 = 1\}$. In terms of polar coordinates, these are given by

$$\xi_1 = \cos \phi \sin \theta; \quad \xi_2 = \sin \phi \sin \theta; \quad \xi_3 = \cos \theta. \quad (2.9)$$

It is straightforward to see, using the definition of Poisson brackets, with $\sigma_1 = \phi, \sigma_2 = \cos \theta$ that

$$\{\xi_i, \xi_j\} = \epsilon_{ijk} \xi_k. \quad (2.10)$$

Since the spherical harmonics, $Y_{lm}(\xi)$ are homogeneous, harmonic polynomials of degree l in the coordinates, ξ_1, ξ_2, ξ_3 , we can expand them in terms of monomials as follows

$$Y_{lm}(\xi_1, \xi_2, \xi_3) = \sum_{i+j+k=l} a_{ijk}^m \xi_1^i \xi_2^j \xi_3^k = \sum_{a_i=1,2,3} c_{a_1, \dots, a_l}^m \xi_{a_1} \cdots \xi_{a_l}, \quad (2.11)$$

with the coefficients, $c_{a_1, \dots, a_l}^m$ being traceless (with respect to any two indices) and totally symmetric tensors. Let us discuss what these properties of the coefficients signify.

1. Totally symmetric means $c_{a_1, \dots, a_l}^m = c_{b_1, \dots, b_l}^m$ for $b_1, \dots, b_l$ being any permutation of $a_1, \dots, a_l$.
2. Traceless over two indices means $c_{a_1, \dots, a_l}^m = 0$. The tracelessness of these tensors is required since the spherical harmonics satisfy the harmonic condition, i.e., $\nabla^2 Y_{lm} = 0$. We can quickly see this for the rank-2 tensor coefficient

$$\begin{aligned}
\nabla^2(c_{ij}^m \xi_i \xi_j) &= c_{ij}^m \frac{\partial}{\partial \xi_\mu} \frac{\partial}{\partial \xi_\mu} (\xi_i \xi_j) \\
&= c_{ij}^m \frac{\partial}{\partial \xi_\mu} [\delta_{i\mu} \xi_j + \delta_{j\mu} \xi_i] \\
&= 2c_{ij}^m [\delta_{i\mu} \delta_{j\mu}] \\
&= 2c_{ii}^m = 0,
\end{aligned}$$

where in the last line we require that c_{ij}^m is traceless.

So far, we have only talked about the algebra of $\text{SDiff}(S^2)$. Now, we try to write the matrix analog of Eq. (2.11), by using the N -dimensional representation of generators of $\text{SO}(3)$, i.e., L_1, L_2, L_3 , such that

$$[L_i, L_j] = i\epsilon_{ijk} L_k. \quad (2.12)$$

Note the similarity of Eq. (2.12) and Eq. (2.10). Next, we define matrix spherical harmonics, $T_{lm}^{(N)}$ by

$$T_{lm}^{(N)} = \sum_{a_i=1,2,3} c_{a_1, \dots, a_l}^m L_{a_1} \cdots L_{a_l}. \quad (2.13)$$

For a given value of N , $T_{lm}^{(N)}$ are independent matrices only for $l = 1, \dots, N-1$. The easiest way to see this is to observe that the number of independent, traceless $N \times N$ matrix is precisely equal to the number of matrices for $l = 1, \dots, N-1$

$$\sum_{l=1}^{N-1} (2l+1) = N^2 - 1. \quad (2.14)$$

In order to make the analogy between spherical harmonics matrix, $T_{lm}^{(N)}$ and spherical harmonics, Y_{lm} more transparent, we list them for $l = 1$ and corresponding $m = 0, \pm 1$.

The cutting-off of higher $l > N$ is analogous to the regularization of divergent integrals in QFT. So, now we have $N^2 - 1$ independent traceless matrices, $T_{lm}^{(N)}$ with the following algebra

$$[T_A^{(N)}, T_B^{(N)}] = i f_{ABC}^{(N)} T_C^{(N)}, \quad (2.15)$$

which is nothing but the Lie algebra of $SU(N)$. Finally, it can be shown that, by rescaling the basis matrices

$$T_A \rightarrow T_A \left(\frac{N^2 - 1}{4} \right)^{\frac{1-l}{2}}, \quad (2.16)$$

(l, m)	Y_{lm}	$T_{lm}^{(N)}$
$(1, -1)$	$\sqrt{\frac{3}{8\pi}}(\xi_1 - i\xi_2)$	$\sqrt{\frac{3}{8\pi}}(L_1 - iL_2)$
$(1, 0)$	$\sqrt{\frac{3}{4\pi}}\xi_3$	$\sqrt{\frac{3}{4\pi}}L_3$
$(1, 1)$	$-\sqrt{\frac{3}{8\pi}}(\xi_1 + i\xi_2)$	$-\sqrt{\frac{3}{8\pi}}(L_1 + iL_2)$

Table 2.1: Spherical harmonics in Cartesian coordinates and their matrix analogs.

we have the following correspondence between the structure constants of $\text{SDiff}(S^2)$ and $SU(N)$ algebras

$$\lim_{N \rightarrow \infty} -if_{ABC}^{(N)} \rightarrow f_{ABC}. \quad (2.17)$$

Thus, we have the following dictionary for mapping of continuum to matrix regularized quantities

$$\{ , \} \rightarrow -i[,], \quad (2.18)$$

$$\int d^2\sigma f \rightarrow \text{Tr} F. \quad (2.19)$$

To complete the discussion, let us summarise the key steps in the procedure. First, we discussed the Lie algebra of APD on S^2 by making use of spherical harmonics. Second, we identified the matrices (L_1, L_2, L_3) in the N -dimensional representation of $SU(2)$ (acting on particles of spin, $s = \frac{N-1}{2}$), whose algebra is similar to that of the Cartesian coordinates (ξ_1, ξ_2, ξ_3) parameterizing 2-sphere. Third, we used the L -matrices to write the matrix analog of the spherical harmonics. The matrices so obtained (followed by a rescaling) turned out to be the generators of $SU(N)$ in the fundamental representation, with structure constants converging to that of $\text{SDiff}(S^2)$ in the large N limit.

Chapter 3

Brief Review of IKKT Matrix Model

In this chapter, we define the IKKT (IIB) matrix model as a large N reduced super Yang-Mills theory and discuss its symmetries. We also outline its relation with the perturbative Type IIB superstring theory.

3.1 The Model

The dimensional reduction of $10D$ maximally supersymmetric Yang-Mills theory, with gauge group $SU(N)$, to a point formally yields the IKKT matrix model, defined by the following action

$$\begin{aligned} S_{IKKT} &= S_b + S_f, \\ S_b &= -\frac{1}{4g^2} \text{tr}[X_\mu, X_\nu][X^\mu, X^\nu], \\ S_f &= -\frac{1}{2g^2} \text{tr}(\bar{\psi} \Gamma^\mu [X_\mu, \psi]). \end{aligned} \tag{3.1}$$

Here, g is the Yang-Mills coupling constant in $(0+0)$ dimensions, X is ten-dimensional vector of $SO(9,1)$ and ψ is the sixteen-component spinor in the Majorana-Weyl representation of $Spin(9,1)$. The bosonic, $X_\mu (\mu = 0, \dots, 9)$ and the fermionic, $\psi_\alpha (\alpha = 1, 2, \dots, 16)$ matrices are $N \times N$ traceless, Hermitian matrices of even and odd Grassmann parity, respectively. The ten gamma matrices, Γ^μ the generators of the Clifford algebra, $Cl(9,1)$. The Lorentz indices are contracted using the Minkowski metric, $\eta = \text{diag}(+, -, -, -)$.

The IKKT model is proposed as a constructive definition of Type IIB superstring theory. To elucidate the connection of the model defined by Eq. (5.1) with the Type IIB superstring theory, we start with the Green-Schwarz action for IIB superstring

written in Schild gauge

$$S_{Schild} = \int d^2\sigma \left[\sqrt{g}\alpha \left(\frac{1}{4}\{X^\mu, X^\nu\}^2 - \frac{i}{2}\bar{\psi}\Gamma^\mu\{X^\mu, \psi\} \right) + \beta\sqrt{g} \right], \quad (3.2)$$

where α, β are constants related to the tension of the string; $\sigma^a = (\tau, \sigma)$, $a = 0, 1$ refers to the worldsheet coordinates ($\sigma \in [0, 2\pi)$); $X^\mu(\tau, \sigma)$, $\mu = 0, 1, \dots, 9$ are the target space coordinates (for a closed string, one can impose the periodic boundary condition, $X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$)¹ and $\{, \}$ is the Poisson bracket on the worldsheet, defined by

$$\{X(\sigma), Y(\sigma)\} = \frac{1}{\sqrt{g}}g^{ab}\partial_a X \partial_b Y, \quad (3.3)$$

with g^{ab} being a worldsheet metric and $g = \det(g^{ab})$.

The matrix analog of this action is given by Goldstone-Hoppe regularization scheme, where we have the following dictionary for interpolation between the continuum and matrix-regularized quantities in the large N limit (with N denoting the size of matrices and acts a regulator as discussed in Eq. (2))

$$\begin{aligned} -i[,] &\longleftrightarrow \{, \}, \\ \text{Tr} &\longleftrightarrow \int d^2\sigma \sqrt{g}. \end{aligned} \quad (3.4)$$

The system defined by Eq. (3.2), after applying Eq. (3.4) becomes

$$S_{Mat}(\beta) = \alpha \left(-\frac{1}{4}\text{Tr}[X_\mu, X_\nu]^2 - \frac{1}{2}\text{Tr}(\bar{\psi}\Gamma^\mu[X_\mu, \psi]) \right) + \beta\text{Tr}\mathbb{1}_N, \quad (3.5)$$

with

$$Z = \sum_{N=0}^{\infty} \int dX d\psi e^{iS_{Mat}(\beta)}. \quad (3.6)$$

Note that the action in Eq. (3.5) is same as Eq. (3.1) up to the βN term. Ishibashi *et al* interpreted β in the matrix-regularized Schild action as the ‘chemical potential’ of the N -eigenvalues of X_μ . The IKKT matrix model, Eq. (3.1) encapsulates the description of an arbitrary number of D-objects (instantons, strings, branes and their mixtures) identified as block-diagonal matrices, with the off-diagonal blocks corresponding to the interactions between these objects. In order to understand how these objects are naturally contained in the model, let’s consider the following background with a block-diagonal form:

$$X_\mu = \begin{pmatrix} X_\mu^{(1)} & 0 \\ 0 & X_\mu^{(2)} \end{pmatrix}. \quad (3.7)$$

¹For the unfamiliar, a worldsheet is the surface traced by a moving string, and the target space is the spacetime in which the string is moving.

It is easy to see that the classical equations of motion for X_μ decouples into equations for the two blocks:

$$[X_\nu^{(1)}, [X^{\mu(1)}, X^{\nu(1)}]] = 0; \quad [X_\nu^{(2)}, [X^{\mu(2)}, X^{\nu(2)}]] = 0, \quad (3.8)$$

which means we have two independent objects, $X_\mu^{(1)}$ and $X_\mu^{(2)}$, which can be further decomposed as:

$$\begin{aligned} X_\mu^{(1)} &= x_\mu^{(1)} \mathbb{1} + \widetilde{X_\mu^{(1)}}; & x_\mu^{(1)} &= \frac{1}{N^{(1)}} \text{tr} (X_\mu^{(1)}); & \text{tr} \left(\widetilde{X_\mu^{(1)}} \right) &= 0; \\ X_\mu^{(2)} &= x_\mu^{(2)} \mathbb{1} + \widetilde{X_\mu^{(2)}}; & x_\mu^{(2)} &= \frac{1}{N^{(2)}} \text{tr} (X_\mu^{(2)}); & \text{tr} \left(\widetilde{X_\mu^{(2)}} \right) &= 0. \end{aligned} \quad (3.9)$$

Here, $x_\mu^{(1)}$ and $x_\mu^{(2)}$ are interpreted as the center of mass of the objects, $X_\mu^{(1)}$ and $X_\mu^{(2)}$ respectively. Generalization to include arbitrary number of objects in the model is straightforward and the presence of many-body configuration is reminiscent of a second quantized theory. Interactions between the blocks can be identified by computing the effective action for these submatrices which requires integrating out the rest of the degrees of freedom. In fact, the β term appears in the one-loop effective action of a single D-string in flat spacetime. This suggests that the matrix regularized Schild action given by Eq. (3.5) describes a low-energy effective theory of the single D-string. The, IKKT matrix model, as mentioned above, encodes multiple objects, instead of a single D-string, which is a strong indication of the model as a constructive definition of Type IIB superstring theory.

3.2 Symmetries of the Model

In this section, we describe the symmetries of the IKKT model. We will make contact with each of these symmetries at some point during the course of this work and so we work out all the details explicitly.

3.2.1 $SU(N)$ Gauge Symmetry

Inherited from the ten-dimensional super Yang-Mills, the model is invariant under the following transformation

$$X_\mu \rightarrow U^\dagger X_\mu U, \quad \psi_\alpha \rightarrow U^\dagger \psi_\alpha U, \quad (3.10)$$

where $U \in \text{SU}(N)$. The bosonic and fermionic parts of the action transforms under Eq. (3.10) as follows

$$\begin{aligned} S_b &\rightarrow S'_b = -\frac{1}{4g^2} \text{tr}[U^\dagger X_\mu U, U^\dagger X_\nu U]^2 \\ &= \frac{1}{4g^2} \text{tr} U^\dagger [X_\mu, X_\nu]^2 U, \\ S_f &\rightarrow S'_f = -\frac{1}{2g^2} \text{tr}(U^\dagger \psi_\alpha U (\mathcal{C}\Gamma^\mu)_{\alpha\beta} U^\dagger [X_\mu, \psi_\beta] U) \\ &= \frac{1}{2g^2} \text{tr}(U^\dagger \bar{\psi} \Gamma^\mu [X_\mu, \psi] U), \end{aligned}$$

where in the first line of S'_f , we used $\bar{\psi} = \psi^T \mathcal{C}$ for a Majorana spinor. The invariance of each of the components of action then follows directly from the basis independent property of trace.

3.2.2 SO(9,1) Lorentz Symmetry

IKKT model has manifest Lorentz symmetry. We will now show that the bosonic and fermionic parts of the action is independently invariant under Lorentz transformation.

Let us first consider the bosonic part of the action. Transformation of the bosonic matrices is given by

$$X^\mu \rightarrow X'^\mu = \Lambda^\mu_\rho X^\rho, \quad (3.11)$$

where Λ^μ_ρ is the Lorentz transformation matrix. This leads to

$$S_b \rightarrow S'_b = -\frac{1}{4g^2} \text{tr} ([X'^\rho, X'^\sigma] \eta_{\rho\mu} \eta_{\sigma\nu} [X'^\mu, X'^\nu]) \quad (3.12)$$

$$= -\frac{1}{4g^2} \text{tr} ([\Lambda^\rho_\alpha X^\alpha, \Lambda^\sigma_\beta X^\beta] \eta_{\rho\mu} \eta_{\sigma\nu} [\Lambda^\mu_\gamma X^\gamma, \Lambda^\nu_\delta X^\delta]) \quad (3.13)$$

$$= -\frac{1}{4g^2} \text{tr} ([X^\alpha, X^\beta] (\Lambda^\rho_\alpha \eta_{\rho\mu} \Lambda^\mu_\gamma) (\Lambda^\sigma_\beta \eta_{\sigma\nu} \Lambda^\nu_\delta) [X^\gamma, X^\delta]) \quad (3.14)$$

$$= -\frac{1}{4g^2} \text{tr} ([X^\alpha, X^\beta] \eta_{\alpha\gamma} \eta_{\beta\delta} [X^\gamma, X^\delta]) \quad (3.15)$$

$$= S_b, \quad (3.16)$$

where in the fourth line, we have used $\Lambda^\rho_\alpha \eta_{\rho\mu} \Lambda^\mu_\gamma = \eta_{\alpha\gamma}$.

Now, moving on to the fermionic part, we work with the infinitesimal version of the Lorentz transformation, which is given by

$$X^\mu \rightarrow (\mathbb{1} + \frac{1}{2} \Omega_{\rho\sigma} M^{\rho\sigma})^\mu_\nu X^\nu. \quad (3.17)$$

Here, $\Omega_{\rho\sigma}$ are the numbers (antisymmetric with respect to the indices), which specify the kind of transformation we are applying, *viz.* rotation or boost. The matrices, $M^{\rho\sigma}$,

are the generators of the transformation with the following entries

$$(M^{\rho\sigma})^\mu{}_\nu = \eta^{\rho\mu}\delta_\nu^\sigma - \eta^{\sigma\mu}\delta_\nu^\rho. \quad (3.18)$$

For transforming spinors, we have to go to the spinorial representation of Lorentz algebra, which is generated by the matrices $S^{\rho\sigma}$ defined as follows

$$S^{\rho\sigma} = \frac{1}{4}[\Gamma^\rho, \Gamma^\sigma] \quad \text{and} \quad (S^{\rho\sigma})^\dagger = -\Gamma^0 S^{\rho\sigma} \Gamma^0, \quad (3.19)$$

where Γ^μ are the usual gamma matrices in the chiral representation. So now we have the following rules for transforming fermions

$$\psi \rightarrow (\mathbb{1} + \frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma})\psi, \quad (3.20)$$

$$\bar{\psi} \rightarrow \bar{\psi}(\mathbb{1} - \frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}). \quad (3.21)$$

Applying Eqs. (3.17), (3.20), and (3.21) to the fermionic part of the action and keeping terms only up to linear order in the infinitesimal parameter, Ω , we get

$$\begin{aligned} S_f \rightarrow S_f + \text{tr} \left(\bar{\psi}\Gamma^\mu \left[\left(\frac{1}{2}\Omega_{\rho\sigma}M^{\rho\sigma} \right)^\nu{}_\mu X_\nu \right] + \bar{\psi}\Gamma^\mu [X_\mu, \left(\frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma} \right)\psi] \right. \\ \left. - \bar{\psi} \left(\frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma} \right) \Gamma^\mu [X_\mu, \psi] \right). \end{aligned} \quad (3.22)$$

Now consider the last two terms inside trace in the above expression, which can be written as (it is easy to see this when the spinors are expressed in component form)

$$\frac{1}{2}\Omega_{\rho\sigma} \left(\bar{\psi}(\Gamma^\mu S^{\rho\sigma})[X_\mu, \psi] - \bar{\psi}(S^{\rho\sigma}\Gamma^\mu)[X_\mu, \psi] \right) = \frac{1}{2}\Omega_{\rho\sigma} \bar{\psi}[\Gamma^\mu, S^{\rho\sigma}][X_\mu, \psi] \quad (3.23)$$

$$= \frac{1}{2}\Omega_{\rho\sigma} \bar{\psi}\Gamma^\mu (M^{\rho\sigma})^\nu{}_\mu [X_\nu, \psi] \quad (3.24)$$

$$= -\frac{1}{2}\Omega_{\rho\sigma} \bar{\psi}\Gamma^\mu (M^{\rho\sigma})_\mu{}^\nu [X_\nu, \psi], \quad (3.25)$$

where in the third line, we have used, $[\Gamma^\mu, S^{\rho\sigma}] = \Gamma^\kappa (M^{\rho\sigma})^\mu{}_\kappa$ with relabelling of the indices and in the last line, the antisymmetry of infinitesimal Lorentz transformation was used. The expression obtained in Eq. (3.25) is just the negative of the first term inside the trace in Eq. (3.22) and thus cancels away and we are left with the original fermionic part of the action. This, combined with the invariance of bosonic part, completes our discussion of Lorentz symmetry of the model.

3.2.3 $\mathcal{N} = 2$ Supersymmetry

The model is invariant under the following supersymmetric transformations

$$\begin{aligned}\delta^{(1)}X_\mu &= i\bar{\epsilon}_1\Gamma_\mu\psi, \\ \delta^{(1)}\psi &= \frac{i}{2}\Gamma^{\mu\nu}[X_\mu, X_\nu]\epsilon_1, \\ \delta^{(2)}X_\mu &= 0, \\ \delta^{(2)}\psi &= \epsilon_2,\end{aligned}$$

where ϵ_1, ϵ_2 are constant Majorana-Weyl spinors parameterizing the supersymmetric transformation. The transformations, $\delta^{(1)}$ are the remnant of $\mathcal{N} = 1$ supersymmetry in ten dimensional Yang-Mills, after dimensional reduction. It is important to note here that due to the dimensional reduction to a point, the commutation relation of the transformation, $\delta^{(1)}$, upto a gauge transformation, is trivial and hence it cannot be regarded as spacetime supersymmetry. On the upside, we get an additional supersymmetry given by $\delta^{(2)}$ and a bosonic symmetry after reduction. The bosonic symmetry reads:

$$\delta X_\mu = c_\mu \mathbb{1} \quad (3.26)$$

If we take the linear combination of $\delta^{(1)}$ and $\delta^{(2)}$ as follows:

$$\tilde{\delta}^{(1)} = \delta^{(1)} + \delta^{(2)}; \quad \tilde{\delta}^{(2)} = i(\delta^{(1)} - \delta^{(2)}) \quad (3.27)$$

then we have the action of the commutator of these transformations as:

$$\begin{aligned}[\tilde{\delta}_\epsilon^{(1)}, \tilde{\delta}_\xi^{(2)}]X_\mu &= 2i\bar{\epsilon}\Gamma_\mu\xi\delta^{ij} \\ [\tilde{\delta}_\epsilon^{(1)}, \tilde{\delta}_\xi^{(2)}]\psi &= 0\end{aligned} \quad (3.28)$$

which is the $\mathcal{N} = 2$ supersymmetry algebra and results in (3.26). Also, since the commutator of two supersymmetry transformations is just a translation, we can identify (3.26) as a spacetime translation if the eigenvalues of the bosonic matrices are interpreted as points in the spacetime, and the transformations, (3.27), then, represent the ten dimensional $\mathcal{N} = 2$ spacetime supersymmetry.

3.3 Euclidean Version

The Euclidean version of the IKKT matrix model is obtained by Wick rotating the Lorentzian version. Under the Wick rotation, we have following transformations

$$X_0 \rightarrow iX_{10}, \quad (3.29)$$

$$\Gamma^0 \rightarrow -i\Gamma^{10}. \quad (3.30)$$

The action is given by Eq. (3.1), where the contractions are now performed using $\delta^{\mu\nu}$ instead of the Minkowski metric. Now, we give an explicit representation of the ten 16×16 gamma matrices (obtained after Weyl projection²) that we will be using in our simulations [Ambjorn 00]

$$\begin{aligned}
\Gamma^1 &= i\sigma^2 \otimes \sigma^2 \otimes \sigma^2 \otimes \sigma^2, \\
\Gamma^2 &= i\sigma^2 \otimes \sigma^2 \otimes \mathbb{1} \otimes \sigma^1, \\
\Gamma^3 &= i\sigma^2 \otimes \sigma^2 \otimes \mathbb{1} \otimes \sigma^3, \\
\Gamma^4 &= i\sigma^2 \otimes \sigma^1 \otimes \sigma^2 \otimes \mathbb{1}, \\
\Gamma^5 &= i\sigma^2 \otimes \sigma^3 \otimes \sigma^2 \otimes \mathbb{1}, \\
\Gamma^6 &= i\sigma^2 \otimes \mathbb{1} \otimes \sigma^1 \otimes \sigma^2, \\
\Gamma^7 &= i\sigma^2 \otimes \mathbb{1} \otimes \sigma^3 \otimes \sigma^2, \\
\Gamma^8 &= i\sigma^1 \otimes \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1}, \\
\Gamma^9 &= i\sigma^3 \otimes \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1}, \\
\Gamma^{10} &= \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1},
\end{aligned} \tag{3.31}$$

where σ^i , $i \in \{1, 2, 3\}$ are the Pauli matrices. In the representation Eq. (3.31), the charge conjugation matrix, $\mathcal{C}$, satisfying

$$\mathcal{C}\Gamma^\mu\mathcal{C}^\dagger = (\Gamma^\mu)^T; \quad \mathcal{C}^T = \mathcal{C}, \tag{3.32}$$

becomes an identity matrix. The $\text{SO}(9, 1)$ Lorentz symmetry of the IKKT model transforms to the $\text{SO}(10)$ rotational symmetry in Euclidean signature.

In order to simulate the IKKT matrix model, we can use the relation between the Yang-Mills coupling (g), number of colors (N) and t'Hooft coupling (λ), given by

$$\lambda = g^2 N \tag{3.33}$$

to write Eq. (3.1) as

$$S_{IKKT} = -\frac{N}{4\lambda} \text{tr}[X_\mu, X_\nu][X^\mu, X^\nu] - \frac{N}{2\lambda} \text{tr}(\bar{\psi}\Gamma^\mu[X_\mu, \psi]). \tag{3.34}$$

Now, the t'Hooft coupling can be absorbed into the field variables by making the following field redefinition

$$X_\mu \rightarrow \lambda^{\frac{1}{4}} X_\mu, \quad \psi \rightarrow \lambda^{\frac{3}{8}} \psi \tag{3.35}$$

²In spacetime with even dimensions, it is always possible to write gamma matrices in off-diagonal block structure. Then Weyl projection amounts to the action of the chirality projection operator defined in a usual sense by the highest-rank gamma matrix.

leading to

$$\begin{aligned}
S_{IKKT} &= S_b + S_f, \\
S_b &= -\frac{N}{4} \text{tr}[X_\mu, X_\nu][X^\mu, X^\nu], \\
S_f &= -\frac{N}{2} \text{tr}(\bar{\psi} \Gamma^\mu [X_\mu, \psi]),
\end{aligned} \tag{3.36}$$

which will be our starting point for applying complex Langevin method to the model.

Convergence of IKKT Matrix Model

The Euclidean IKKT matrix model is well defined without any cutoff, unlike the Lorentzian IKKT, which requires infrared cutoff [Kim 12a]. The convergence of partition function for both supersymmetric and bosonic Euclidean IKKT model were shown by Austing and Wheeler [Austing 01b, Austing 01a], which can be summarized as follows:

- **Supersymmetric:** Convergent for $D = 4, 6$ and 10 .
- **Bosonic:** Convergent for $D \geq 3$ for gauge group $\text{SU}(N)$ with $N \geq 4$.

Chapter 4

Complex Langevin Method

The methods to evade the sign-problem induced by complex actions rely on the simple idea of complexifying the field variables when the measure is complex, so that the complex measure is non-negative, or fluctuating only mildly, thereby reducing the severity of the sign problem. In this chapter, we provide a quick introduction to one of these methods, *viz.* complex Langevin dynamics for dealing with the sign problem, in a completely general setup. The application of CLM to the matrix models is left for the next chapter.

4.1 Real Langevin

Real Langevin in the context of field theories, known as stochastic quantization, was developed by Parisi [Parisi 81]. It describes stochastic evolution of the field variables along a fictitious time t , governed by the following equation

$$\frac{d\phi}{dt} = -\frac{\delta S[\phi]}{\delta\phi} + \eta(t), \quad (4.1)$$

for some field, ϕ , where $S'[\phi]$ is the drift term coming from the action functional, $S[\phi]$ and is the classical part of the evolution, whereas $\eta(t)$ is the Gaussian noise encoding quantum fluctuations around the classical part, obeying the following statistics

$$\langle\eta(t)\rangle_\eta = 0; \quad \langle\eta(t)\eta(t')\rangle_\eta = 2\delta(t-t'). \quad (4.2)$$

Note the η in the subscript means that we are averaging over the noise. The stochastic evolution generates fields at time t , with a probability density, $P(\phi, t)$, which evolves according to the Fokker-Planck equation

$$\frac{\partial P(\phi, t)}{\partial t} = \frac{\delta}{\delta\phi(t)} \left(\frac{\delta}{\delta\phi(t)} + \frac{\delta S[\phi]}{\delta\phi(t)} \right) P(\phi, t). \quad (4.3)$$

The expectation of an observable, $\mathcal{O}$ in the framework of Langevin dynamics is given by

$$\langle \mathcal{O}(\phi) \rangle_\eta = \int d\phi P(\phi, t) \mathcal{O}(\phi). \quad (4.4)$$

The justification of the method relies on whether the large time limit of $P(\phi, t)$ asymptotes to $e^{-S[\phi]}$ or not, in other words, whether the field configurations are distributed according to the Boltzmann weights or not. For real Langevin, indeed, it can be shown that

$$\lim_{t \rightarrow \infty} P(\phi, t) = e^{-S[\phi]}, \quad (4.5)$$

by assuming *ergodicity*, which refers to the equivalence of ensemble average and time average. Before going on to the complex extension, we write the simulation-friendly version of Eq. (4.1), i.e., discretized form of Langevin evolution, which reads

$$\phi_{n+1} = \phi_n - \epsilon S'[\phi_n] + \sqrt{\epsilon} \eta_n, \quad (4.6)$$

where Langevin time is discretized as $\theta = n\epsilon$, n and ϵ being the time step and step size, respectively. In practice, the expectation value of observables are estimated by taking the statistical average over time T as follows

$$\langle \mathcal{O} \rangle_\eta \equiv \frac{1}{T} \sum_{\text{th}}^{\text{th}+T} \mathcal{O}(\phi_\eta), \quad (4.7)$$

where 'th' is the thermalization time for the system after which the field configurations or the observables fluctuate around the thermal equilibrium.

4.2 Complex Langevin

Idea, Challenges, and Resurgence

Parisi [Parisi 83] and Klauder [Klauder 84] independently postulated complex Langevin dynamics as a formal extension of real Langevin process for systems with complex action. The real configuration space becomes a complex space, i.e., $\phi \rightarrow \phi_R + i\phi_I$, the observables are continued to holomorphic functions, $\mathcal{O}(\phi_R + i\phi_I)$ and the stochastic evolution is given by

$$\begin{aligned} \frac{d\phi_R}{dt} &= -\text{Re} \left(\frac{\delta S[\phi]}{\delta \phi} \right) + \eta_R(t), \\ \frac{d\phi_I}{dt} &= -\text{Im} \left(\frac{\delta S[\phi]}{\delta \phi} \right) + \eta_I(t), \end{aligned} \quad (4.8)$$

where the noises, $\eta_R(t), \eta_I(t)$ obey ¹

$$\begin{aligned}\langle \eta_R(t) \rangle &= \langle \eta_I(t) \rangle = 0, \\ \langle \eta_R(t) \eta_R(t') \rangle &= 2N_R \delta(t - t'), \\ \langle \eta_I(t) \eta_I(t') \rangle &= 2N_I \delta(t - t'), \\ N_R - N_I &= 1.\end{aligned}\tag{4.9}$$

The discretization of Eq. (4.8) is straightforward and is given by: (relabeling $\phi_R \rightarrow x, \phi_I \rightarrow y$)

$$x_{n+1} = x_n - \epsilon \operatorname{Re}(S'(x_n, y_n)) + \sqrt{\epsilon} \eta_n, \tag{4.10}$$

$$y_{n+1} = y_n - \epsilon \operatorname{Im}(S'(x_n, y_n)). \tag{4.11}$$

In complex Langevin, the evolution of positive probability density, $P(\phi_R, \phi_I, t)$ defined on complex space is governed by the Fokker-Planck equation

$$\frac{\partial P(\phi_R, \phi_I; t)}{\partial t} = \frac{\delta}{\delta \phi_R(t)} \left(\frac{\delta}{\delta \phi_R(t)} + \frac{\delta S[\phi]}{\delta \phi_R} \right) + \frac{\delta}{\delta \phi_I(t)} \frac{\delta S[\phi]}{\delta \phi_I}. \tag{4.12}$$

On the other hand, evolution of the complex density, $\rho(\phi, t)$ on real space is described by Eq. (4.3). We can write expectation of a holomorphic observable $\mathcal{O}$ in real and complex configuration space as follows

$$\langle \mathcal{O} \rangle_{P(t)} = \int \mathcal{O}(\phi_R + i\phi_I) P(\phi_R, \phi_I; t) d\phi_R d\phi_I; \quad \langle \mathcal{O} \rangle_{\rho(t)} = \int \mathcal{O}(\phi) \rho(\phi, t) d\phi. \tag{4.13}$$

Complex Langevin is justified if $\langle \mathcal{O} \rangle_{P(t)} = \langle \mathcal{O} \rangle_{\rho(t)}$ holds, given they agree at $t = 0$. This has not been proved yet in full generality, but certain conditions, known as correctness criteria, have been identified which justifies the usage of CLM and ensures the validity of results obtained. We will discuss more about the correctness criteria in Chapter 5.

Soon after the inception of complex Langevin, it was tried to tackle the systems with sign-problem, but the results were mostly mixed which led to loss of interest in the method. Sometimes the method was successful; at other times it simply failed, due to non-convergence and runaway trajectories. Worse yet, in some cases, it converged to wrong limit. The interest in CLM was rekindled by the works of J. Berges and I.O. Stamatescu [Berges 05] on simulations of quantum fields in real time. Not much later, Aarts and Stamatescu successfully applied complex Langevin to QCD models with finite chemical potential [Aarts 08], which suffers from sign problem. The resurgence

¹Generally, the noise is taken to be real, i.e., $N_I = 0$ to limit the excursions in imaginary directions and for our purposes, we will also take real noise.

has led to a deeper understanding of several shortcomings of the method encountered in the earlier studies and formulation of optimization methods to overcome them, consequently, allowing the exploration of several systems with manifest sign-problem. In particular, usage of adaptive step size is shown to avoid runaway trajectories. Unlike a fixed stepsize, the adaptive stepsize is taken to be inversely proportional to the drift term, i.e., larger steps are made in the simulation when it is safe to do so. While discussing the consistency of complex Langevin by the equivalence of the two expectation values in Eq. (4.13), one of the crucial detail that was swept under the rug is the sufficiently fast decay of the probability of the drift term at large magnitude and near the poles of the drift term. In order to cure this issue, a controlled exploration of the complex plane by fields is required. Gauge cooling for localised excursion have proven to be successful in restoring the required decay of distribution of drift terms. We will discuss the appearance of large excursions and singular drift problem in the context of IKKT matrix model in the next chapter.

Chapter 5

Applying CLM to Euclidean IKKT Matrix Model

In this chapter, we discuss the application of CLM to simulate the IKKT matrix model. We will also go through the occurrence of various technicalities mentioned in the previous chapter, in the context of matrix models and possible ways to overcome them. We also discuss the observables to investigate the SSB of rotational symmetry.

The partition function for the IKKT model in Euclidean signature reads

$$Z = \int dX d\psi e^{-(S_b + S_f)}, \quad (5.1)$$

where S_b and S_f are given by Eq. (3.36). After integrating out the fermions using Berezin integration, one obtains the Pfaffian of the fermionic operator

$$Z = \int dX e^{-S_b} \text{Pf}(\mathcal{M}), \quad (5.2)$$

where $\mathcal{M}$ is a $16(N^2 - 1) \times 16(N^2 - 1)$ antisymmetric matrix, which represents the following linear transformation

$$\psi_\alpha \rightarrow (\mathcal{M}\psi)_\alpha \equiv (\Gamma_\mu)_{\alpha\beta} [X_\mu, \psi_\beta]. \quad (5.3)$$

Substituting Eq. (5.2) in Eq. (5.1) we have

$$Z = \int dX e^{-S_{\text{eff}}}; \quad S_{\text{eff}} = S_b - \log \text{Pf}(\mathcal{M}). \quad (5.4)$$

For matrix models, we can consider the evolution of all the elements of the bosonic matrices in the fictitious Langevin time, governed by the following equation

$$\frac{d(X_\mu)_{ij}}{dt} = -\frac{\partial S_{\text{eff}}}{\partial (X_\mu)_{ji}} + (\eta_\mu)_{ij}(t), \quad (5.5)$$

where $\eta_\mu(t)$ belongs to the Gaussian Orthogonal Ensemble (GOE), with the joint probability distribution of the elements given by $\propto \exp(-\frac{1}{4} \int \text{tr}(\eta_\mu(t))^2)$ (for details,

see Appendix A). Notice the swapping of indices in the drift term as compared to the term on the left hand side, which happens when one tries to take derivative of a number (action is a number) with respect to a matrix. More precisely, we have the identity, $\frac{\partial \text{Tr } AB}{\partial A} = B^T$.

The drift term can be written as

$$\frac{\partial S_{\text{eff}}}{\partial (X_\mu)_{ji}} = \frac{\partial S_b}{\partial (X_\mu)_{ji}} - \frac{\partial \log \text{Pf}(\mathcal{M})}{\partial (X_\mu)_{ji}}. \quad (5.6)$$

We will derive the explicit form of the gradients in the next section. Now, the discretized analog of Eq. (5.5) is given by

$$(X_\mu)_{ij}(t + \Delta t) = (X_\mu)_{ij}(t) - \Delta t \frac{\partial S_{\text{eff}}}{\partial (X_\mu)_{ji}(t)} + \sqrt{\Delta t} (\eta_\mu)_{ij}(t). \quad (5.7)$$

The step size, Δt is chosen to be adaptive for a stabilized evolution and a faster convergence towards the expectation value [Aarts 10]. Due to the complex nature of the action, the bosonic matrices transforms into a general traceless matrix from Hermitian traceless. Now we move on to the computation of drift term.

5.1 Gradient Computations

In this section, we work out the expressions for bosonic and fermionic parts of the gradient in full detail.

5.1.1 Gradient of the Bosonic Action

The bosonic part of the action is given by

$$S_b = -\frac{1}{4} N \text{tr} [X_\mu, X_\nu]^2. \quad (5.8)$$

We have

$$\frac{\partial S_b}{\partial (X_\mu)_{ji}} = -N(2(X_\nu X_\mu X_\nu) - X_\mu X_\nu X_\nu - X_\nu X_\nu X_\mu)_{ij} \quad (5.9)$$

$$= -N[X_\nu, [X_\mu, X_\nu]]_{ij}. \quad (5.10)$$

5.1.2 Gradient of Fermionic Action

The $N \times N$ bosonic and fermionic matrices, are traceless and Hermitian, which can be expanded in terms of the generators, t^a of $\text{SU}(N)$

$$X_\mu = \sum_{a=1}^{N^2-1} X_\mu^a t^a \quad \text{and} \quad \psi_\alpha = \sum_{b=1}^{N^2-1} \psi_\alpha^b t^b, \quad (5.11)$$

where A_μ^a are the real numbers, ψ_α^b are the Grassmann numbers and the generators are normalized as $\text{tr}(t_a t_b) = \delta_{ab}$. Putting the expansions Eq. (5.11) in the fermionic action, one obtains

$$S_f = -\frac{1}{2}N \text{tr} (\psi_\alpha^a t^a \Gamma_{\alpha\beta}^\mu [X_\mu^c t^c, \psi_\beta^b t^b]) \quad (5.12)$$

$$= -\frac{1}{2}N \psi_\alpha^a \Gamma_{\alpha\beta}^\mu X_\mu^c \psi_\beta^b \text{tr} (t^a [t^c, t^b]) \quad (5.13)$$

$$= -\frac{iN}{2} \psi_\alpha^a \Gamma_{\alpha\beta}^\mu X_\mu^c \psi_\beta^b f^{cbd} \text{tr} (t^a t^d) \quad (\text{using } [t^c, t^b] = i f^{cbd} t^d) \quad (5.14)$$

$$= -\frac{iN}{2} \psi_\alpha^a (\Gamma_{\alpha\beta}^\mu X_\mu^c f^{cba}) \psi_\beta^b \quad (\text{using } \text{tr}(t^a t^d) = \delta^{ad}) \quad (5.15)$$

$$= \psi_\alpha^a \left(\frac{iN}{2} \Gamma_{\alpha\beta}^\mu X_\mu^c f^{abc} \right) \psi_\beta^b = \psi_\alpha^a \mathcal{M}_{\alpha a, \beta b} \psi_\beta^b. \quad (5.16)$$

Here the fermionic operator can be written in a more illustrative form, using the properties of structure constants of $\text{SU}(N)$ as follows

$$\mathcal{M}_{\alpha a, \beta b} = \frac{iN}{2} \Gamma_{\alpha\beta}^\mu X_\mu^c f^{abc} \quad (5.17)$$

$$= \frac{1}{2} N \Gamma_{\alpha\beta}^\mu X_\mu^c \text{tr} (t^a [t^b, t^c]) \quad (5.18)$$

$$= \frac{1}{2} N \Gamma_{\alpha\beta}^\mu X_\mu^c \text{tr} (t^c [t^a, t^b]) \quad (5.19)$$

$$= \frac{1}{2} N \Gamma_{\alpha\beta}^\mu \text{tr} (X_\mu^c t^c [t^a, t^b]) \quad (5.20)$$

$$= \frac{1}{2} N \Gamma_{\alpha\beta}^\mu \text{tr} (X_\mu [t^a, t^b]). \quad (5.21)$$

With the explicit matrix form of the fermionic operator, it is now straightforward to take the derivative of $\mathcal{M}$ to get

$$\frac{\partial \mathcal{M}_{\alpha a, \beta b}}{\partial (X_\mu)_{ji}} = \frac{N}{2} \Gamma_{\alpha\beta}^\mu ([t^a, t^b])_{ij}. \quad (5.22)$$

The complete fermionic gradient to be used in CLM becomes

$$\begin{aligned} \frac{\partial S_f}{\partial (X_\mu)_{ji}} &= -\frac{\partial \log \text{Pf}(\mathcal{M})}{\partial (X_\mu)_{ji}} \\ &= -\frac{1}{2} \text{tr} \left(\mathcal{M}^{-1} \frac{\partial \mathcal{M}}{\partial (X_\mu)_{ji}} \right). \end{aligned} \quad (5.23)$$

An Aside: Computational Efficiency

From the fermionic part of the drift term, we see the requirement of computing the inverse of the matrix, $\mathcal{M}$ explicitly. Worse yet, the size of this matrix is huge, even for reasonable values of N . This is computationally the most intensive part, and naturally, the bottleneck for the time complexity of the simulation. For values up to $N = 4$, we

could perform simulations of full IKKT model in reasonable time. But for larger N , we need a more efficient way to compute the gradient. Looking the term again, we realise that trace of a big matrix is required in the end. It turns out that a computationally less alternative to compute the trace explicitly is to get an approximate result using Stochastic trace estimation. See Appendix B for details.

Correctness Criteria

As mentioned in the review of complex Langevin in Chapter (4), unlike the case with real action, we lack a rigorous proof that the probability distribution of the fields configuration, $\rho(X_\mu, \tau)$ asymptotes to the Boltzmann weight, $e^{-S_{\text{eff}}[X_\mu]}$ in the limit $\tau \rightarrow \infty$. Therefore, we can not know *a priori* whether the results obtained from CLM are reliable or not and hence, a correctness criteria is required. In Ref. [Nagata 16a], it was argued that a sufficient condition for correct convergence is the exponential or faster fall off of the probability distribution of the magnitude of drift, u , defined by

$$u = \sqrt{\frac{1}{10N^2} \sum_{\mu=1}^{10} \sum_{i,j=1}^N \left| \frac{\partial S_{\text{eff}}[X_\mu]}{\partial (X_\mu)_{ij}} \right|^2}. \quad (5.24)$$

For our works, we monitor the probability distribution of u extensively to ascertain the reliability of the results. This fall-off condition of $p(u)$ is violated either when the field configurations (X_μ) wander too far in the enlarged space (due to the complexification of fields), or spending too much time near the singularities of the drift term. We now turn to the appearance of these problems in the context of IKKT matrix model and the ways to overcome them.

5.2 Singular Drift Problem

In the gradient for fermionic part, given by Eq. (5.23), we see the presence of $\mathcal{M}^{-1}$ coming from taking the derivative of $\log \text{Pf}(\mathcal{M})$. Due to this, the singular drift problem occurs when the eigenvalues of the fermionic operator accumulate too close to zero. In order to cure this, adding deformation terms has been suggested in Ref. [Ito 16] where they applied it to investigate SSB patterns in the $\text{SO}(4)$ -symmetric matrix model and found promising results. This method was also successfully applied to the $6D$ version of Euclidean IKKT matrix model [Anagnostopoulos 18]. The addition of deformation term shifts the eigenvalues of $\mathcal{M}$ away from the origin, thus restoring the required fall-off of $p(u)$. Now, we describe the type of deformation we will be using for our purpose. The simplest one is the gaussian deformation, a very general form of which

can be written as follows

$$\Delta S = \frac{N}{2}\epsilon \sum_{\mu=1}^{10} m_{\mu} \text{tr} X_{\mu}^2 + \frac{N}{2} \sum_{\alpha,\beta=1}^{16} \text{tr}(\psi_{\alpha}(\mathcal{C}\mathcal{A})_{\alpha\beta}\psi_{\beta}), \quad (5.25)$$

where $\mathcal{A}$ is a complex 16×16 antisymmetric matrix and $\mathcal{C}$ is the charge conjugation matrix. Now, given the fact that the set formed by $\{\mathbb{1}, \Gamma^{\mu}, \Gamma^{\mu\nu}, \dots, \Gamma^{\ast}\}$ (where $\Gamma^{\ast}$ is the highest rank gamma matrix) is a basis for a general complex matrix of appropriate dimensions. For $D = 10$, Majorana-Weyl spinors severely constrains the allowed rank of gamma matrices, i.e., only the bilinears with rank three and seven tensor survives (which are equivalent due to the duality relations), see Ref. [Wetterich 83]. The matrix $\mathcal{A}$, then takes the following form

$$\mathcal{A} = \frac{i}{3!} \sum_{\mu,\nu,\rho=1}^{10} m_{\mu\nu\rho} (\Gamma^{\mu}\Gamma^{\nu}\Gamma^{\rho}), \quad (5.26)$$

where $m_{\mu\nu\rho}$ is a 3-form. Furthermore, we pick $m_{\mu\nu\rho} = m_f \epsilon_{\mu\nu\rho}$ for $\mu, \nu, \rho \in \{8, 9, 10\}$ and zero otherwise with $\epsilon_{\mu\nu\rho}$ being the Levi-Civita symbol in three dimensions. This type of deformation breaks the $\text{SO}(10)$ rotational symmetry to $\text{SO}(7) \times \text{SO}(3)$. For the mass vector, m_{μ} in the bosonic deformation, we make the following choice

$$m_{\mu} = (0.5, 0.5, 0.5, 1, 2, 4, 8, 8, 8, 8). \quad (5.27)$$

Let us see the effect of these deformations on the drift plots (in the semi-log setup) obtained for different values of m_f and ϵ for $N = 2$.

From Fig. 5.1, it is clear that the deformations are indeed necessary, as the drift plot corresponding to undeformed IKKT Model ($m_f = 0.0, \epsilon = 0.0$) does not possess the required fall-off. As we increase the bosonic deformation parameter, ϵ , we see some improvements but the decay follows a power-law behaviour rather than exponential or faster, which again is not sufficient to validate the results obtained from CLM. Now we turn on the fermionic deformation parameter, m_f . In Fig. 5.2, we observe that for $\epsilon = 0.0$, the correctness criteria is not satisfied. For $\epsilon = 0.1$, there is a power law type decay behaviour. For $\epsilon = 0.2$ and higher, we have the required fall-off of the probability distribution to ensure the reliability of the results.

5.3 Excursion Problem and Gauge Cooling

As discussed in Chapter (3), the fermionic part of the action is complex in the Euclidean signature, and the model is said to suffer from the notorious *sign-problem*. The very complex nature of the action leads to excursion of bosonic matrices, X_{μ} into

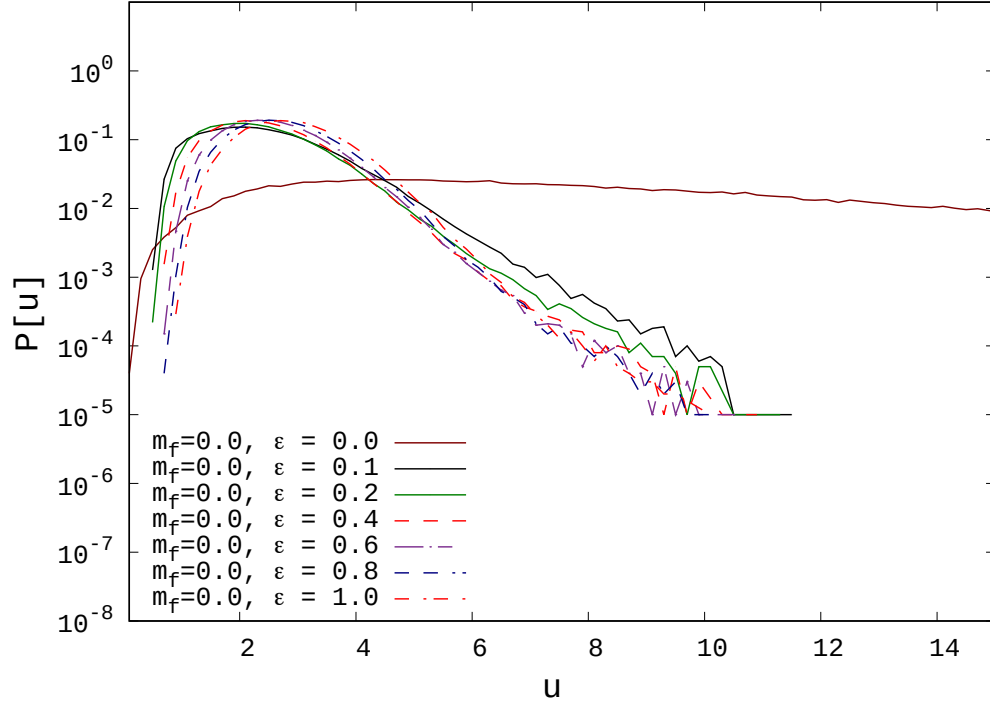


Figure 5.1: The drift plots of $N = 2$ for various ϵ and $m_f = 0$.

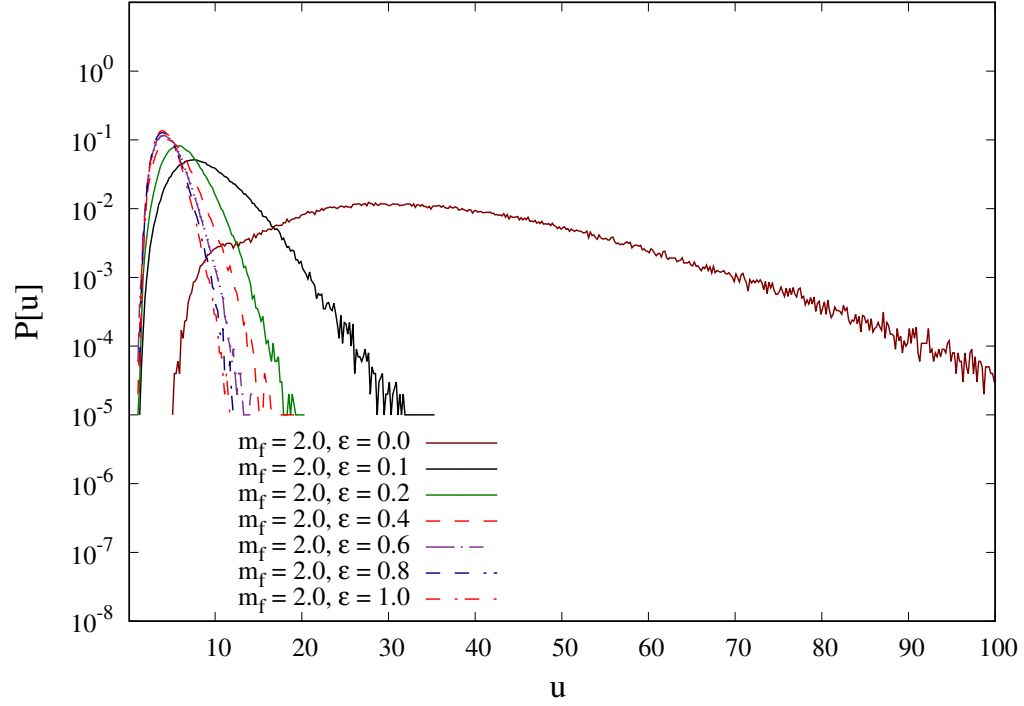


Figure 5.2: The drift plots of $N = 2$ for various ϵ and $m_f = 2$.

the imaginary directions, *viz.* from $SU(N)$ to the enlarged group, $SL(N, \mathbb{C})$. These movements must be controlled to ensure the validity of results obtained by complex Langevin simulations as mentioned in the last section. Gauge cooling, proposed in Ref. [Seiler 13], is a method which makes it possible to limit the wide venture of the matrices into the imaginary directions. In order to measure the extent of excursion of the matrices in $SL(N, \mathbb{C})$, we use ‘Hermiticity norm,’ $\mathcal{N}_H$, which, as the name suggests, refers to the deviation of the matrix from Hermiticity. It is given by

$$\mathcal{N}_H = \frac{1}{10N} \sum_{\mu=1}^{10} \text{tr} \left((X_\mu - X_\mu^\dagger) (X_\mu^\dagger - X_\mu) \right). \quad (5.28)$$

This is zero for Hermitian matrices, and non-zero otherwise. Upon complexification, the gauge group of the model also enhances from $SU(N)$ to $SL(N, \mathbb{C})$, i.e., observables and drift term are invariant under the transformation, $X_\mu \rightarrow g X_\mu g^{-1}$ for $g \in SL(N, \mathbb{C})$. It is important to note, however that $\mathcal{N}_H$ is not invariant under the above mentioned gauge transformation, and we exploit this property to minimize the deviation from Hermiticity. In particular, we use gradient descent method for minimization. For computing the gradient of Eq. (5.28), consider the following infinitesimal gauge transformation

$$g = 1 + \epsilon_a t_a, \quad (5.29)$$

with t_a being the generators of $SU(N)$ and ϵ_a is real for $SL(N, \mathbb{C})$ transformation. The fields change as (keeping only terms linear in ϵ_a)

$$X_\mu \rightarrow g X_\mu g^{-1} = (1 + \epsilon_a t_a) X_\mu (1 - \epsilon_a t_a) = X_\mu + \epsilon_a [t_a, X_\mu], \quad (5.30)$$

$$X_\mu^\dagger \rightarrow (g^{-1})^\dagger X_\mu^\dagger g^\dagger = (1 - \epsilon_a t_a) X_\mu^\dagger (1 + \epsilon_a t_a) = X_\mu^\dagger - \epsilon_a [t_a, X_\mu^\dagger]. \quad (5.31)$$

Substituting this back to Eq. (5.28) gives the required gradient (again keeping terms only up to $\mathcal{O}(\epsilon)$)

$$\Delta \mathcal{N}_H = \epsilon_a f_a, \text{ where } f_a = \frac{1}{N} \sum_{\mu} \text{tr} (t_a [X_\mu, X_\mu^\dagger]). \quad (5.32)$$

To summarise, the process of gauge cooling then amounts to applying the following transformation with $g = e^{-\alpha f_a t_a}$ at each Langevin step successively until the norm stops decreasing

$$\begin{aligned} \tilde{X}_\mu(t) &= g X_\mu(t) g^{-1}, \\ X_\mu(t + \Delta t) &= \tilde{X}_\mu(t) + \Delta t \frac{\partial S}{\partial \tilde{X}_\mu(t)} + \sqrt{\Delta t} \eta(t). \end{aligned} \quad (5.33)$$

Here, f_a is the gradient as computed in Eq. (5.32) and α is a real positive parameter chosen to minimize the norm. The gauge cooling parameter α can be chosen a constant

or adaptive. The adaptive version is given by

$$\alpha_{\text{ad}} = \frac{\epsilon\alpha}{\Sigma_a|f_a| + D_0}, \quad (5.34)$$

for some suitable D_0, α and Langevin step size, ϵ . Adaptive gauge cooling leads to very fast evolution towards Hermiticity. Gauge cooling has been formally justified in Ref. [Nagata 16b]. In Fig. 5.3, we show the time evolution of Hermiticity norm for some N values. We find that at least for these N , we do not need to apply gauge cooling, since the deviations are well under control. Note, however that the deviation has increased significantly on increasing N , and it is clear that gauge cooling must be applied for larger N to get sensible results.

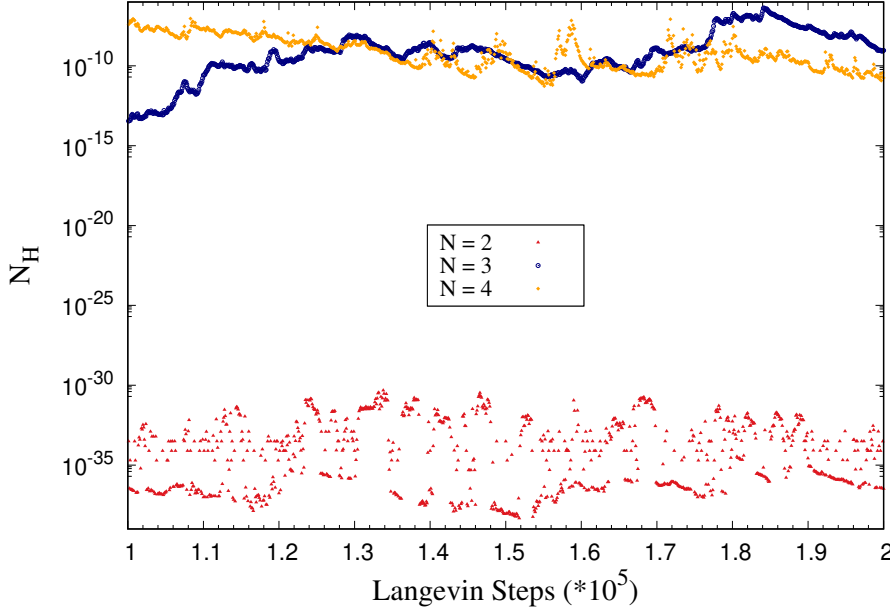


Figure 5.3: Langevin time history of $\mathcal{N}_H$ for $N = 2, 3, 4$ with $\epsilon = 0.4, m_f = 2.0$.

We show the drift plots in Fig. 5.4 to see if the necessary fall-off condition is satisfied without gauge cooling, and it is obvious from figure below that indeed the results obtained for these N can be trusted.

5.4 Observables

Now having discussed some of the technicalities of using CLM, we return to the question of observing spontaneous breaking of $\text{SO}(10)$ rotational symmetry. To this end, we will follow the steps in Ref. [Anagnostopoulos 20]. As an order parameter, we consider the 'radial extent' of eigenvalues denoted by $\langle \lambda_\mu \rangle$ of the bosonic matrices,

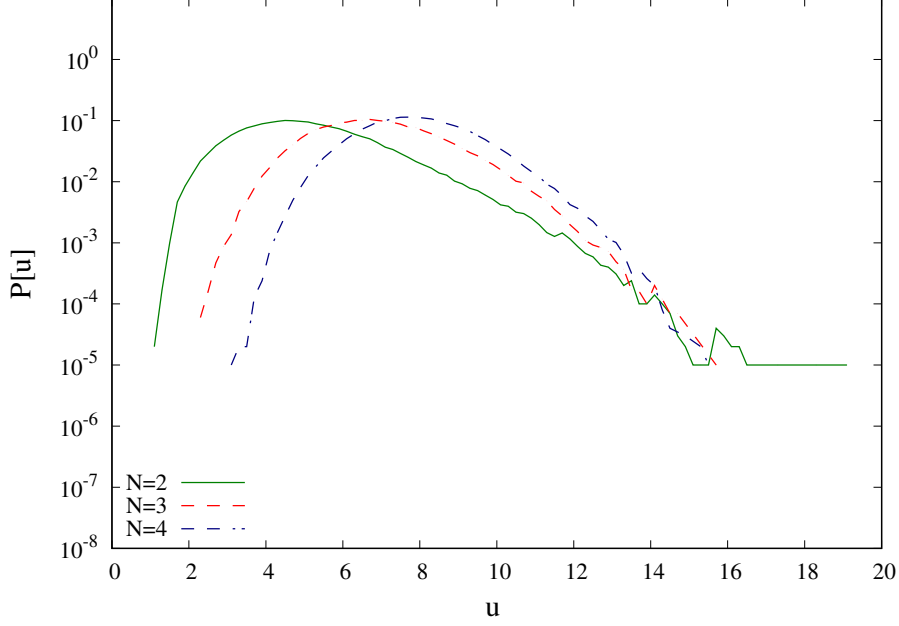


Figure 5.4: Drift plots for $N = 2, 3, 4$ with $\epsilon = 0.4, m_f = 2.0$.

which is defined as

$$\langle \lambda_\mu \rangle = \left\langle \frac{1}{N} \text{Tr} X_\mu^2 \right\rangle. \quad (5.35)$$

In the IKKT matrix model, since we interpret the eigenvalues of the bosonic matrices as points in the spacetime, these extents are analogous to the radii of emergent spacetime along different dimensions. As discussed in Sec. 5.2, we have two parameters, the fermionic deformation, m_f and ϵ for the bosonic part. In Table 6.2, we summarize the symmetry of the model depending on whether the deformations are zero or non-zero for the choice of mass vector given by Eq. (5.27).

Parameters	Symmetry
$\epsilon \neq 0, m_f \neq 0$	$SO(3) \times SO(3)$
$\epsilon = 0, m_f \neq 0$	$SO(7) \times SO(3)$
$\epsilon = 0, m_f = 0$	$SO(10)$

Table 5.1: Symmetries of the deformed model.

Given the deformations, in order to observe spontaneous symmetry breaking, the methodology can now be succinctly described as follows

1. For $m_f \neq 0$, we first make large N extrapolation and then $\epsilon \rightarrow 0$ which restores the symmetry of the model up to $SO(7) \times SO(3)$. Then we can investigate if the $SO(7)$ symmetry further breaks into smaller subgroups by measuring the expectation of extent values.

2. After large N and $\epsilon \rightarrow 0$ extrapolation, we take the limit $m_f \rightarrow 0$ which corresponds to undeformed IKKT and investigate the SSB of $\text{SO}(10)$ rotational symmetry.

After the required extrapolations, if all of the extents are not equivalent, i.e., grow along some dimensions, $d < 10$, and shrink along others, we say that the $\text{SO}(10)$ rotational symmetry spontaneously breaks to $\text{SO}(d)$. And naturally, the dimensions with larger extents will be the macroscopic ones of the vacuum so obtained. It is in this sense that we say one can predict the dimensionality of emergent spacetime by solving the IKKT model.

Chapter 6

Results

In this chapter, we discuss the results obtained from the simulations of IKKT using complex Langevin. For bosonic case, we could easily go upto $N = 64$. For the full model, given the time and the capabilities of the computing resources available, at present, we could only go up to $N = 4$ with reasonable computation time.

6.1 Bosonic IKKT Model

We have shown the Langevin time history of extents along some directions in Fig. 6.1 and the action itself in Fig. 6.2 where we observe excellent agreement of simulation results for the action with the analytical result given in Ref. [Hotta 99b]. It is clear from the figures that the observables have thermalized. Moreover, from the plot of action, we see that it is real and the complex part comes entirely from the Pfaffian, which is why the bosonic IKKT model does not suffer from the sign problem.

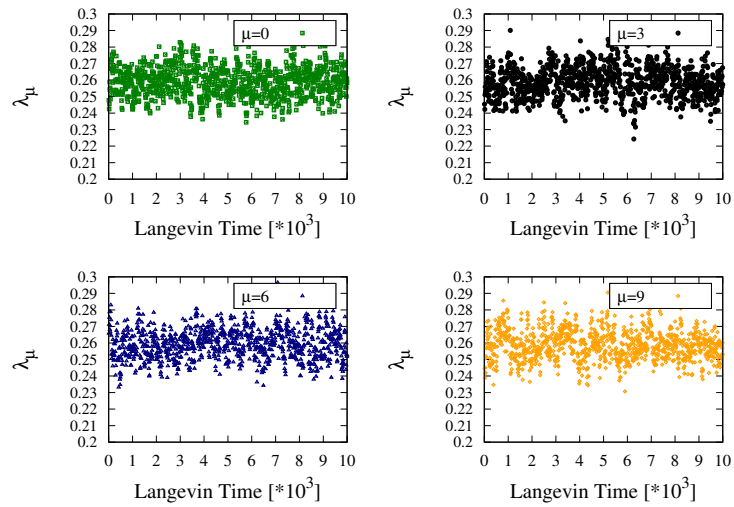


Figure 6.1: Langevin time history of $\lambda_0, \lambda_3, \lambda_6$ and λ_9 for $N = 64$.

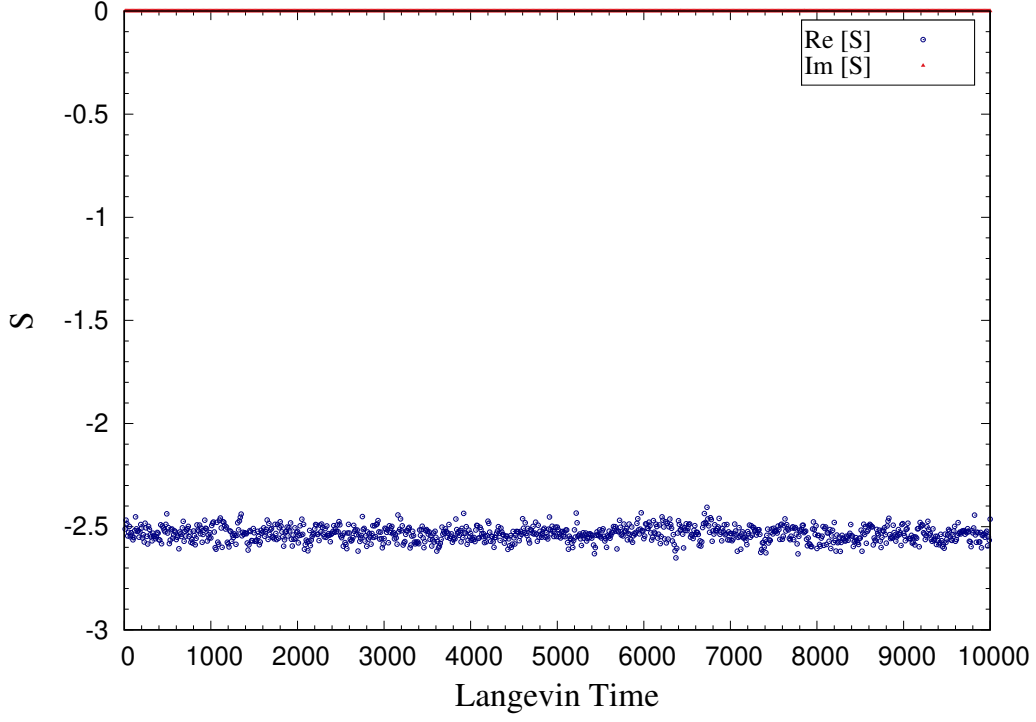


Figure 6.2: Langevin time history of S for $N = 64$.

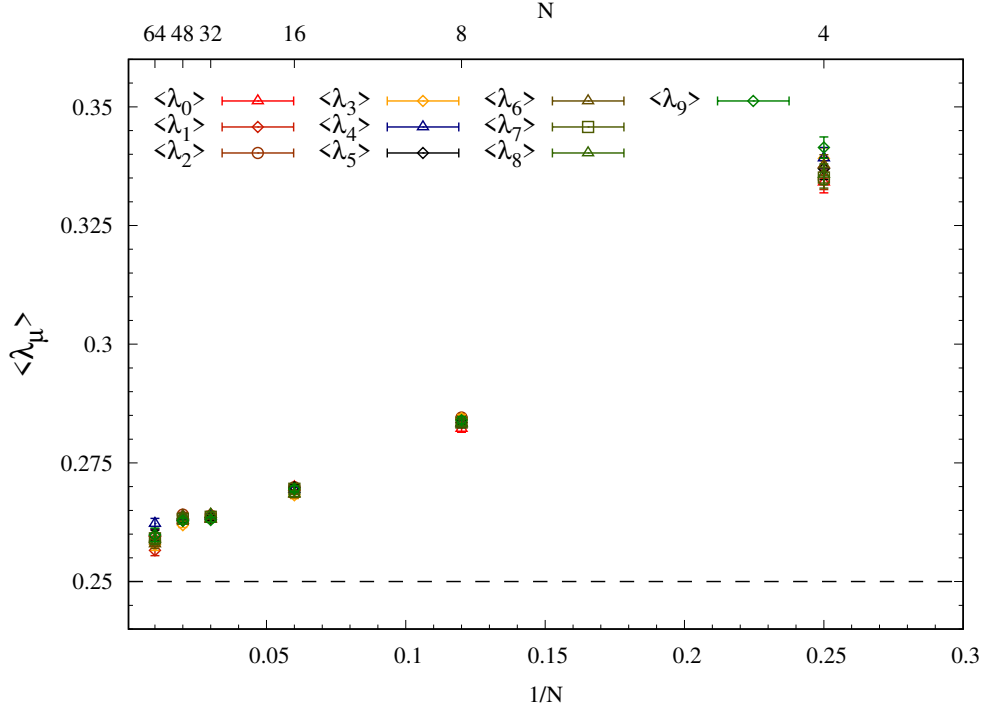


Figure 6.3: The expectation value of the extent of space-time $\langle \lambda_\mu \rangle$ for the bosonic IKKT model are plotted against $1/N$. The dashed line represents the analytical large N result [Hotta 99b].

The expectation value of the extents is shown in Fig. 6.3. It is clear that all the eigenvalues are equal in value and thus SSB is absent. We show the probability distributions of the drift term in Fig. 6.4 and observe the required fall-off condition, thereby ensuring the credibility of the results.

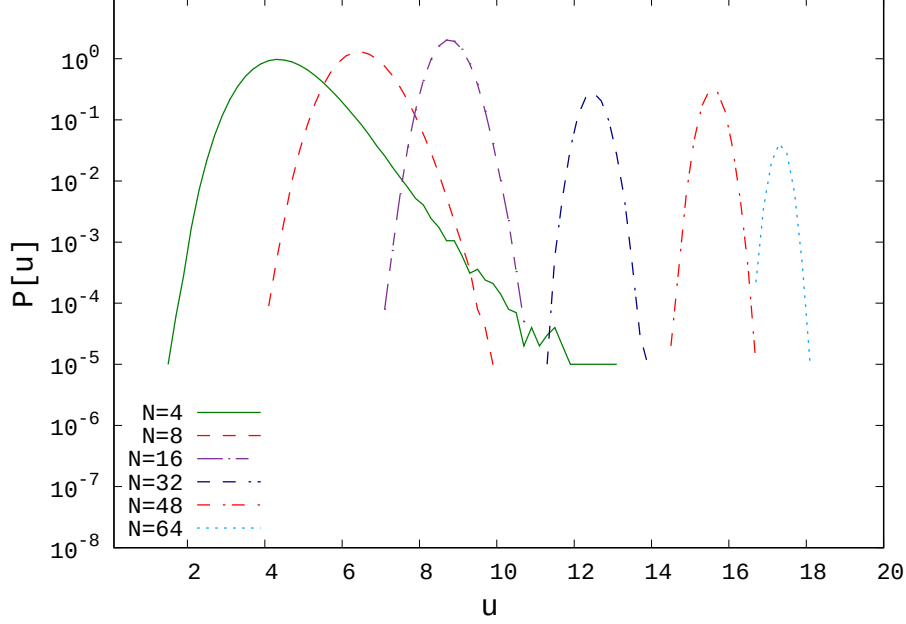


Figure 6.4: Probability distribution, $p(u)$ of the magnitude of drift term, u is plotted in semi-log scale for various N .

6.2 Supersymmetric IKKT Model

For the full model, we define the observable, $\rho_\mu(\epsilon, m_f)$ for each value of (ϵ, m_f) as a normalized extent, defined by

$$\rho_\mu(\epsilon, m_f) = \frac{\langle \lambda_\mu \rangle_{\epsilon, m_f}}{\sum_{\nu=1}^{10} \langle \lambda_\nu \rangle_{\epsilon, m_f}}. \quad (6.1)$$

The reason for this is the fact that $\lambda \propto \frac{1}{\epsilon}$, which makes the extrapolation of $\epsilon \rightarrow 0$ a bit difficult. Moreover, the increase in the extents as $\epsilon \rightarrow 0$ may give us a false impression of SSB. The normalized extents remove this dependence and are better order parameters to consider for investigating SSB. Also, it is expected and was observed that the ρ_μ values are almost identical along the dimensions with equal mass value, which means for (5.27), we can take the average of ρ_1, ρ_2, ρ_3 to represent extent corresponding to $m = 0.5$ and similarly the average of $\rho_8, \rho_9, \rho_{10}$ for $m = 8$. In Fig. (6.5), we show that the observables have thermalized.

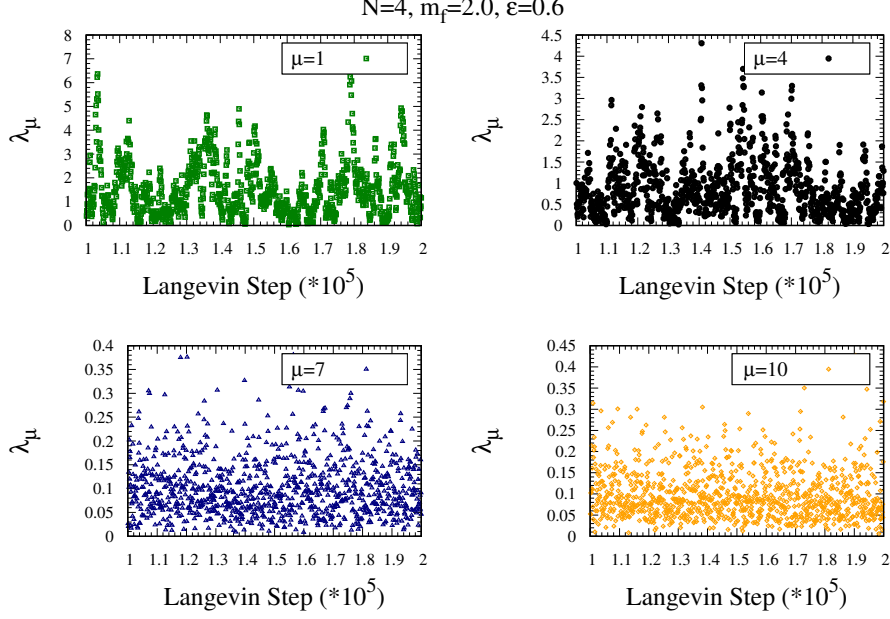


Figure 6.5: Langevin time history of $\lambda_1, \lambda_4, \lambda_7$ and λ_{10} for $N = 4, m_f = 2.0, \epsilon = 0.6$.

Here, since we could not go beyond $N = 4$, we are considering it as a large N -limit for our computations. Figs. (6.6), (6.7), and (6.8) show the variation in the extents as $\epsilon \rightarrow 0$ for $m_f = 3, 2$, and 1 , respectively. The curves represent the best fit to the values obtained by CLM using a quadratic fitting function. In the limit $\epsilon \rightarrow 0$, if there is no SSB, one should obtain ρ_μ to be equal for the group $\mu = 1, 2, \dots, 7$ and the group $\mu = 8, 9, 10$. It is clear from the $\epsilon \rightarrow 0$ extrapolation that even for N as small as 4 , we can see that the the extents for the above-mentioned groups are not equivalent. Further, in Fig. (6.9), which represents the variation in $\lim_{\epsilon \rightarrow 0} \rho_\mu$ as $m_f \rightarrow 0$, we see that there is no restoration of $\text{SO}(10)$ symmetry. At this stage, we do not make claim as to which subgroup the $\text{SO}(10)$ symmetry breaks into since it requires simulations at larger values of N , but the growth and shrinkage of extents along some directions are reminiscent of SSB and we hope to report it for large N . We show the corresponding drift plots in Figs. (6.10), (6.11), and (6.12), which ensures that the results obtained can be trusted.

In tables (6.1) and (6.2), we show the $\epsilon \rightarrow 0$ and $m_f \rightarrow 0$ extrapolated values, respectively, of the normalized extents. Note, that for $m_f \rightarrow 0$ extrapolations in table (6.2), we do not have errors, because we are fitting three points ($m_f = 3.0, 2.0, 1.0$) by a quadratic function of the form $a + bm_f + cm_f^2$, which has three parameters and thus all the errors are zero by definition.

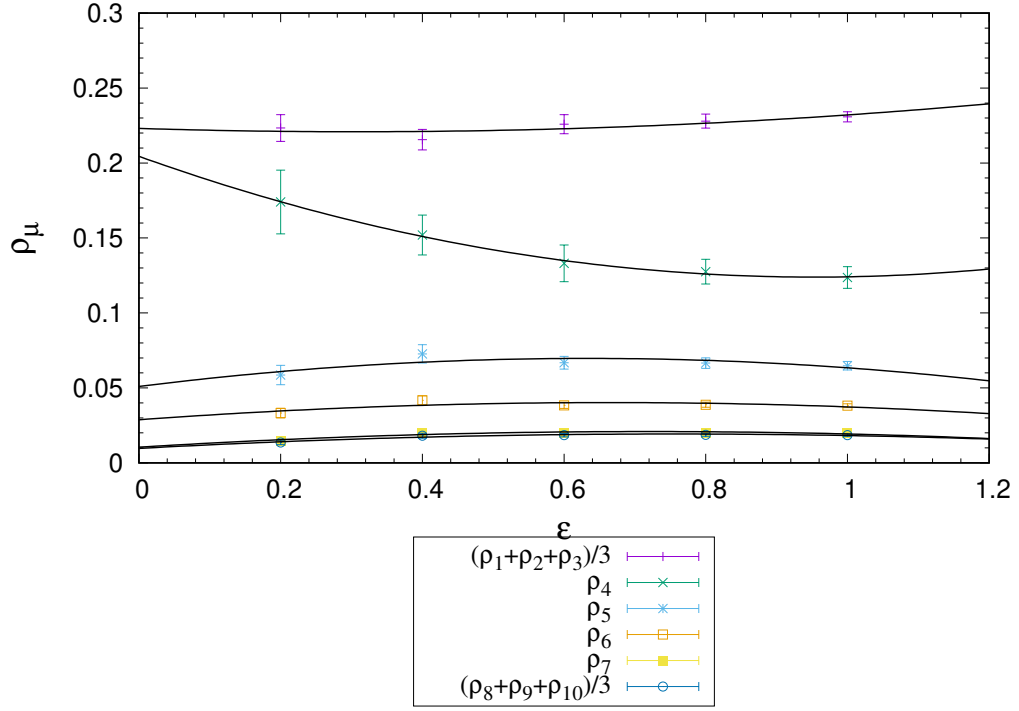


Figure 6.6: The expectation value of normalized extent of space-time $\langle \rho_\mu \rangle$ for the full IKKT model are plotted against ϵ with $m_f = 3.0$.

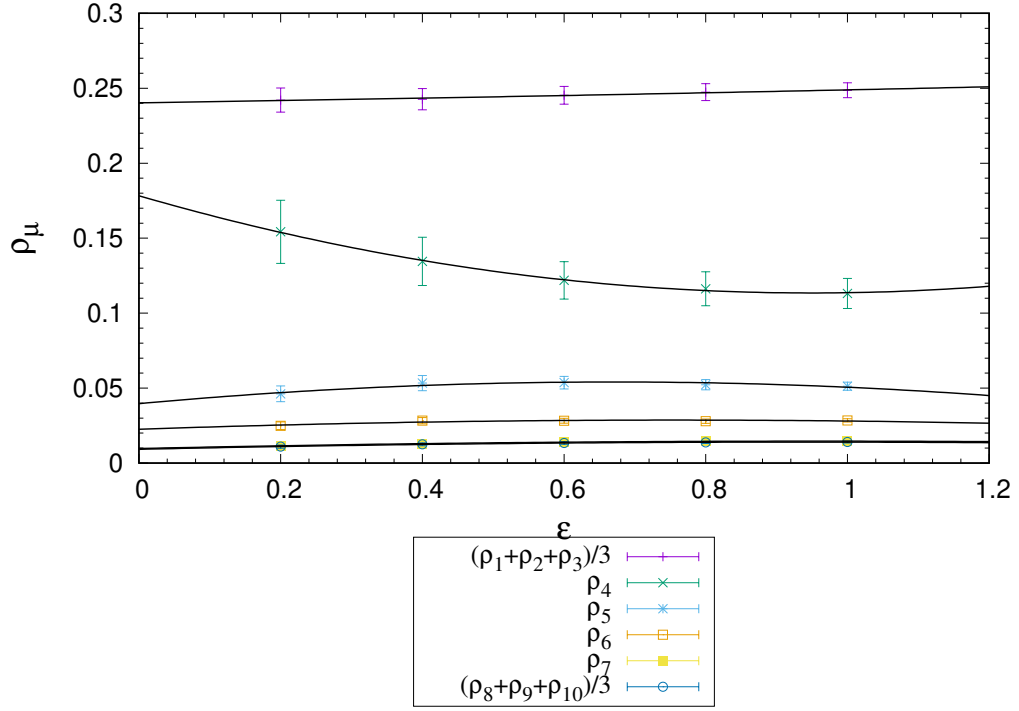


Figure 6.7: The expectation value of normalized extent of space-time $\langle \rho_\mu \rangle$ for the full IKKT model are plotted against ϵ with $m_f = 2.0$.

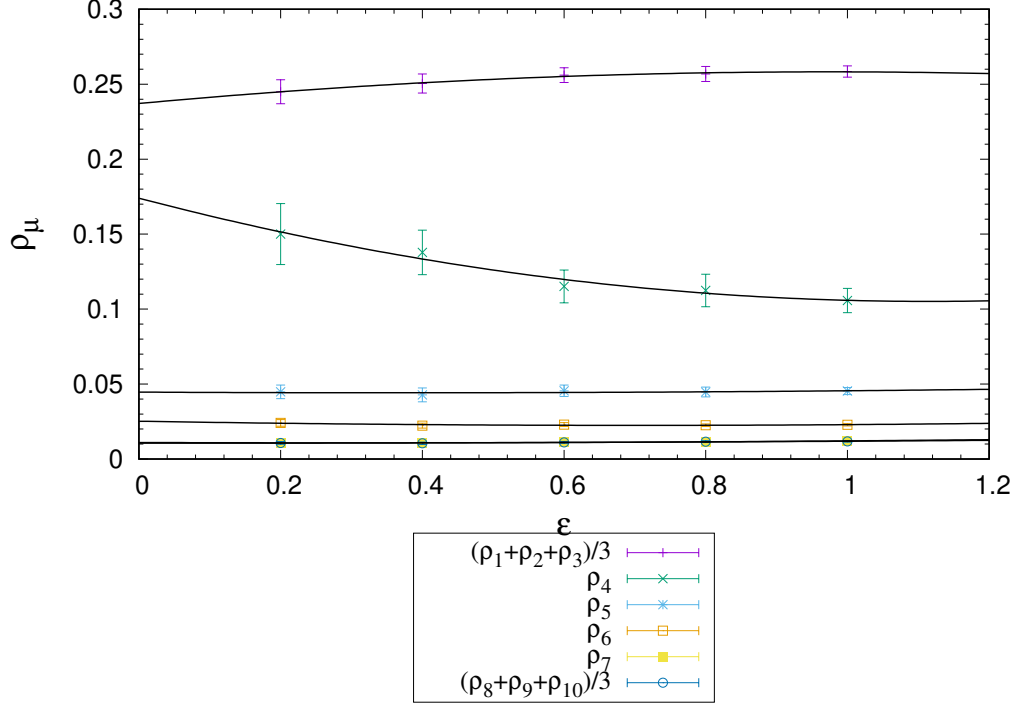


Figure 6.8: The expectation value of normalized extent of space-time $\langle \rho_\mu \rangle$ for the full IKKT model are plotted against ϵ with $m_f = 1.0$.

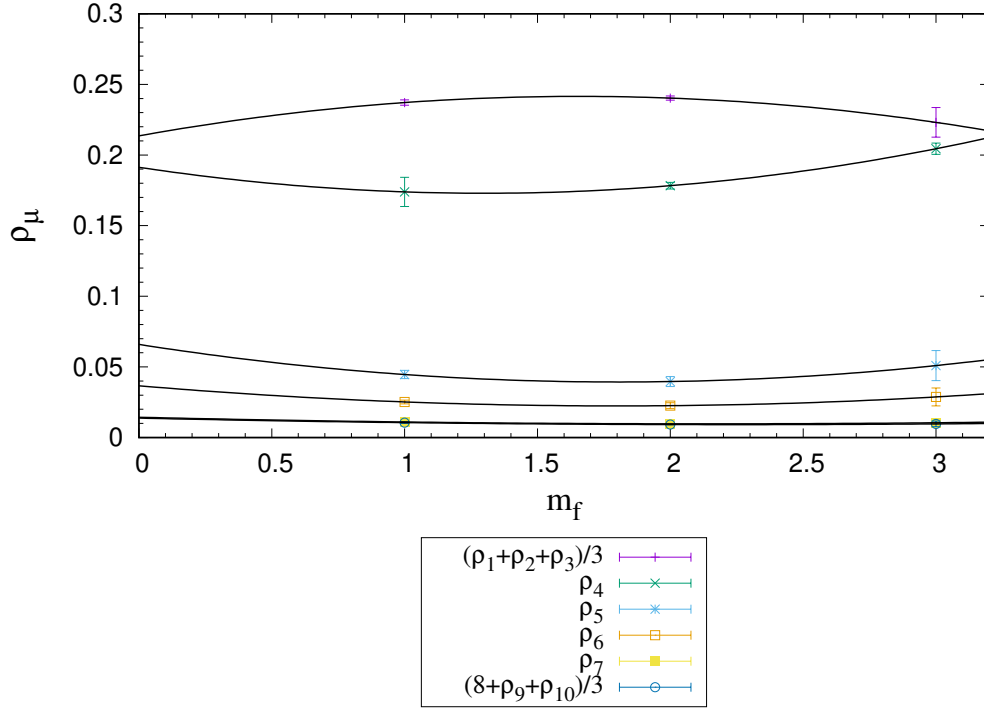


Figure 6.9: The $\epsilon \rightarrow 0$ extrapolated values of the normalized extents, $\lim_{\epsilon \rightarrow 0} \langle \rho_\mu \rangle$ are plotted against m_f to obtain the $m_f \rightarrow 0$ limit. The curves represent the best fit to the values using a quadratic fitting function.

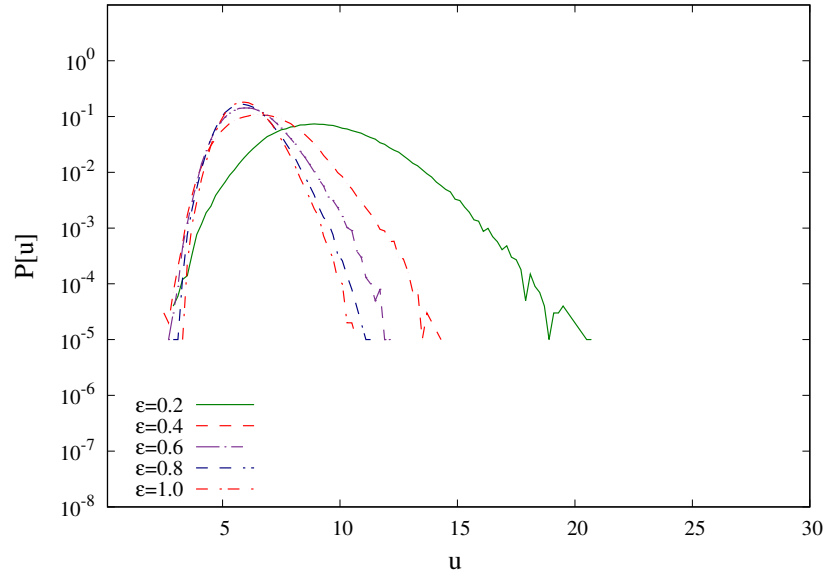


Figure 6.10: Drift plots for $N = 4, m_f = 3.0$ and various ϵ values.

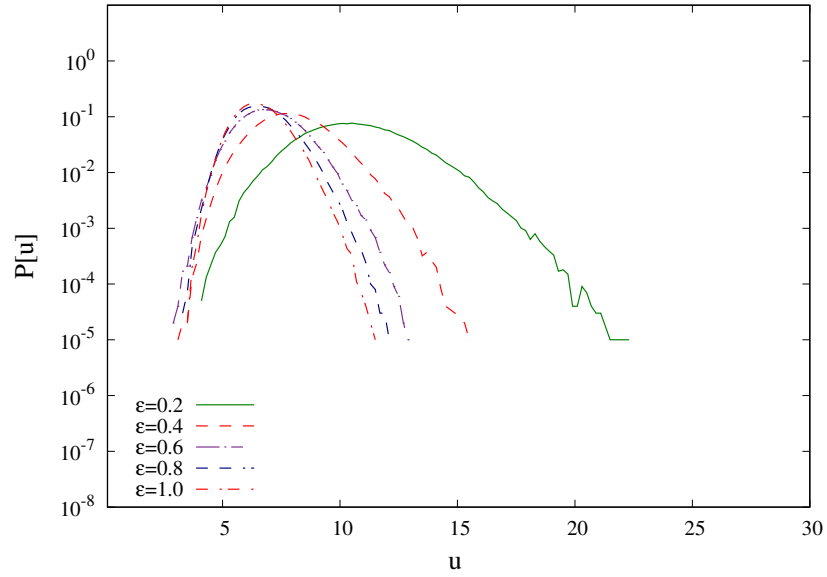


Figure 6.11: Drift plots for $N = 4, m_f = 2.0$ and various ϵ values.

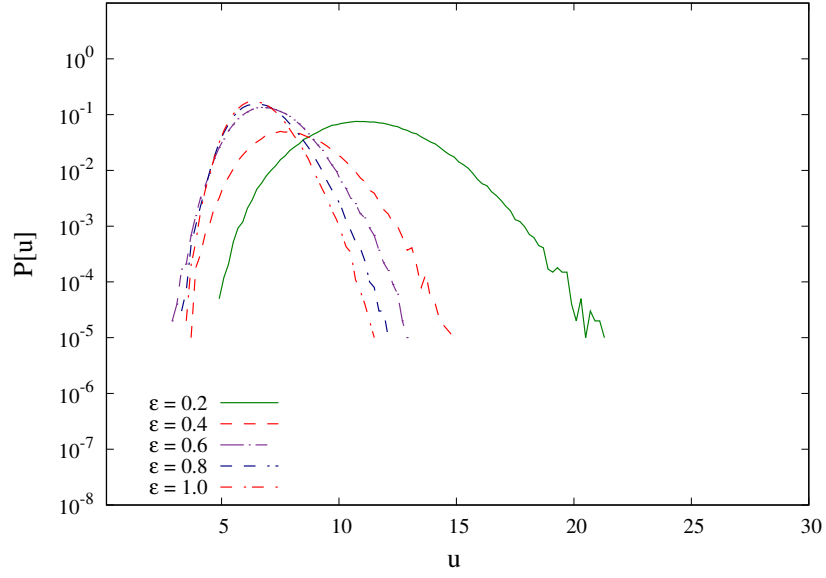


Figure 6.12: Drift plots for $N = 4, m_f = 1.0$ and various ϵ values.

m_f	ρ_μ	Extrapolated Value
3.0	$(\rho_1 + \rho_2 + \rho_3)/3$	0.223089(10490)
	ρ_4	0.204455(4022)
	ρ_5	0.05095(1066)
	ρ_6	0.028739(6366)
	ρ_7	0.010469(2658)
	$(\rho_8 + \rho_9 + \rho_{10})/3$	0.009653(1972)
2.0	$(\rho_1 + \rho_2 + \rho_3)/3$	0.240307(1453)
	ρ_4	0.178286(2338)
	ρ_5	0.039632(3529)
	ρ_6	0.022563(2103)
	ρ_7	0.009720(275)
	$(\rho_8 + \rho_9 + \rho_{10})/3$	0.009196(241)
1.0	$(\rho_1 + \rho_2 + \rho_3)/3$	0.237126(1933)
	ρ_4	0.173987(10340)
	ρ_5	0.044628(2892)
	ρ_6	0.025200(1322)
	ρ_7	0.011016(709)
	$(\rho_8 + \rho_9 + \rho_{10})/3$	0.010565(589)

Table 6.1: $\epsilon \rightarrow 0$ extrapolation of the normalized extents for various m_f .

ρ_μ	Extrapolated Value
$(\rho_1 + \rho_2 + \rho_3)/3$	0.213546
ρ_4	0.191318
ρ_5	0.065937
ρ_6	0.036650
ρ_7	0.014359
$(\rho_8 + \rho_9 + \rho_{10})/3$	0.013760

Table 6.2: $m_f \rightarrow 0$ extrapolation of the normalized extents, ρ_μ .

Chapter 7

Summary and Future Prospects

In this thesis, we studied both the bosonic and supersymmetric versions of the IKKT matrix model in Euclidean signature using Complex Langevin method. The goal was to investigate spontaneous breaking of $SO(10)$ rotational symmetry. For the bosonic IKKT matrix model, we did not observe SSB in the large N limit, which supports the claim that Pfaffian plays a crucial role in the SSB. The simulations for full IKKT model was done up to $N = 4$ for this thesis and we have observed that the all the extents are not equivalent, which indicates the presence of SSB. For $N = 4$, we found the presence of singular drift problem for $m_f < 1.0$ and $\epsilon < 0.2$. As discussed in the Appendix of [Anagnostopoulos 20], the severity of singular drift problem decreases for larger values of N , which might allow us to simulate for lower values of (m_f, ϵ) in our future work leading to better extrapolations see if we get the SSB reported in their paper.

We have also identified a deformed IKKT matrix model, which we believe has the essential deformation terms to overcome the technicalities posed by complex Langevin Method. In addition to that, there is a Myers term which preserves half of supersymmetry, i.e., there are 16 supercharges in the model. Due to preservation of SUSY, arbitrary mass vector for bosonic deformation is not allowed, and there is only one parameter instead of the two parameters here, (m_f, ϵ) . Consequently, the three step extrapolation here reduces to only two step extrapolation. Following [Bonelli 02], the deformed model is given by

$$\begin{aligned} S &= S_{IKKT} + S_m; \quad \text{where,} \\ S_{IKKT} &= N \text{tr} \left(-\frac{1}{4} [X_\mu, X_\nu] [X^\mu, X^\nu] - \frac{1}{2} (\bar{\psi} \Gamma^\mu [X_\mu, \psi]) \right), \\ S_m &= N \text{tr} \left(\frac{i}{8} \bar{\psi} N_3 \psi - M^{\mu\nu} X_\mu X_\nu + i N^{\mu\nu\rho} X_\mu [X_\nu, X_\rho] \right). \end{aligned} \tag{7.1}$$

Here, $N^{\mu\nu\rho}$ is totally antisymmetric and $N_3 = \Gamma^{\mu\nu\rho} N_{\mu\nu\rho}$ and the last term in

S_m is the Myers term. The action is invariant under the following supersymmetric transformation

$$\delta X^\mu = \frac{1}{2}\bar{\epsilon}\Gamma^\mu\psi \quad \text{and} \quad \delta\psi = \frac{1}{4}[X^\mu, X^\nu]\Gamma_{\mu\nu}\epsilon - \frac{i}{16}X^\mu(\Gamma_\mu N_3 + 2N_3\Gamma^\mu)\epsilon, \quad (7.2)$$

provided

$$[N_3(\Gamma_\mu N_3 + 2N_3\Gamma^\mu) + 4^3 M^{\mu\nu}\Gamma_\nu]\epsilon = 0. \quad (7.3)$$

It is easy to see that $N_3 = \mu\Gamma^{8910}$ and $M = -\frac{\mu^2}{4^3}(\mathbb{1}_7 \oplus 3\mathbb{1}_3)$ solves (7.3). In S_m , we see that the first two terms are the ones we were using as deformation terms in our works. For $\mu \neq 0$, the $SO(10)$ symmetry breaks to $SO(7) \times SO(3)$, and for $\mu = 0$, we get undeformed IKKT matrix model. To sum up, we can investigate SSB of IKKT matrix model by first taking the large N limit and then making the $\mu \rightarrow 0$ extrapolation.

Appendix A

Gaussian Ensembles

In this appendix, we collect some of the important results from random matrices which are relevant in the context of matrix models. We start with describing the most common ensemble, GOE, which we have used for noise matrix in the complex Langevin dynamics.

Definition A.0.1. Let A be a $N \times N$ real and symmetric matrix with independently and identically distributed Gaussian random variables. If the diagonal entries, $A_{ii} \sim \mathcal{N}(0, 2)$ and $A_{ij} \sim \mathcal{N}(0, 1)$ for $i \neq j$, then A belongs to the **Gaussian Orthogonal Ensemble**.

Using this definition, let us now try to compute the joint probability distribution (pdf) of entries of GOE, $p(A)$. Since the entries are independent, the joint pdf is given by the product of distribution of the entries

$$p(A) = \prod_{i=1}^N \frac{1}{\sqrt{4\pi}} e^{-\frac{1}{4}A_{ii}^2} \prod_{1 \leq i < j \leq N} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}A_{ij}^2} \quad (\text{A.1})$$

$$= \left(\frac{1}{\sqrt{4\pi}}\right)^N \left(\frac{1}{\sqrt{2\pi}}\right)^{\frac{N(N-1)}{2}} e^{-\frac{1}{4}[\sum_{i=1}^N A_{ii}^2 + 2\sum_{1 \leq i < j \leq N} A_{ij}^2]} \quad (\text{A.2})$$

$$= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2\pi}\right)^{\frac{N(N+1)}{4}} e^{-\frac{1}{4}[\sum_{i=1}^N A_{ii}^2 + \sum_{i \neq j} A_{ij}^2]} \quad (\text{A.3})$$

$$= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2\pi}\right)^{\frac{N(N+1)}{4}} e^{-\frac{1}{4}[\sum_{i,j=1}^N A_{ij}^2]} \quad (\text{A.4})$$

$$= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2\pi}\right)^{\frac{N(N+1)}{4}} e^{-\frac{1}{4}[\sum_{i,j=1}^N A_{ij} A_{ji}]} \quad (\text{using } A_{ij} = A_{ji}) \quad (\text{A.5})$$

$$= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2\pi}\right)^{\frac{N(N+1)}{4}} e^{-\frac{1}{4}[\sum_{i=1}^N A_{ii}^2]} \quad (\text{A.6})$$

$$= \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{2\pi}\right)^{\frac{N(N+1)}{4}} e^{-\frac{1}{4}\text{Tr } A^2}. \quad (\text{A.7})$$

Equation A.7 is what we have for the probability distribution of the noise matrix. As an aside, it is important to note that the term ‘orthogonal’ in the ensemble does not mean that A is orthogonal, rather it signifies the invariance of probability distribution under the transformation, $A \rightarrow O^T A O$ for some orthogonal matrix O .

The other common ensemble is Gaussian Unitary Ensemble (GUE), which can be defined as

Definition A.0.2. Let A be a $N \times N$ complex Hermitian matrix with independently and identically distributed Gaussian random variables. If the diagonal entries, $A_{ii} \sim \mathcal{N}(0, 1)$ (real) and $A_{ij} \sim \mathcal{N}(0, \frac{1}{2}) + i\mathcal{N}(0, \frac{1}{2})$ for $i \neq j$, then A belongs to the **Gaussian Unitary Ensemble**.

The joint pdf of GUE, up to the normalization factor is given by $e^{-\frac{1}{2}\text{Tr } A^2}$. Again, the term ‘unitary’ refers to the invariance of probability distribution of A under unitary transformations.

The significance of these ensembles lies in the fact that it is possible to describe the distribution of their eigenvalues explicitly. The joint pdf of the eigenvalues, up to the normalization factor, can be written collectively for different Gaussian ensembles by making use of the Dyson Index β as follows

$$p(\lambda_1, \lambda_2, \dots, \lambda_N) \propto \prod_{1 \leq i < j \leq N} |\lambda_i - \lambda_j|^\beta \prod_{j=1}^N e^{-\frac{\beta}{4}\lambda_j^2}, \quad (\text{A.8})$$

with $\beta = 1(2)$ for GOE (GUE).

Appendix B

Stochastic Trace Estimation

In this appendix, we discuss an efficient technique to compute the gradient of the fermionic action, which is computationally the most intensive task in the Langevin evolution.

B.1 $\text{Tr}(M)$

In this section we demonstrate the stochastic estimation of the trace of a matrix. Let us consider a general $n \times n$ matrix M . We need an n -dimensional white-noise vector η such that its elements obey the following property

$$\langle \eta_a^* \eta_b \rangle = \delta_{ab}, \quad (\text{B.1})$$

where $a, b = \{1, 2, 3, \dots, n\}$.

In order to implement the above property we consider

$$\eta_a = \frac{\alpha_a + i\beta_a}{\sqrt{2}} \quad \text{and} \quad \eta_a^* = \frac{\alpha_a - i\beta_a}{\sqrt{2}}, \quad (\text{B.2})$$

where α_a and β_a obey the normal distribution $N(0, 1)$, that is, $\langle \alpha_a^2 \rangle = \langle \beta_a^2 \rangle = 1$. Now for the trace computation we have

$$\text{Tr } M = \sum_{b=1}^n M_{bb} \quad (\text{B.3})$$

$$= \sum_{a,b=1}^n M_{ab} \delta_{ab} \quad (\text{B.4})$$

$$= \sum_{a,b=1}^n M_{ab} \langle \eta_a^* \eta_b \rangle \quad (\text{B.5})$$

$$= \sum_{a,b=1}^n \langle \eta_a^* M_{ab} \eta_b \rangle. \quad (\text{B.6})$$

Hence to compute the trace of the matrix M we can compute the quantity $\langle \eta_a^* M_{ab} \eta_b \rangle$.

B.2 $\text{Tr} \left[\frac{\partial \mathcal{M}}{\partial (A_\mu)_{ji}} \mathcal{M}^{-1} \right]$

The fermion part of the drift requires computation of the trace of a quantity involving the inverse of fermion matrix. The fermion matrix $\mathcal{M}$ has the size $16(N^2-1) \times 16(N^2-1)$ and the direct calculation of $\mathcal{M}^{-1}$ has the time complexity of $O(N^6)$. Using noisy-estimator method we can reduce the time complexity to $O(N^4)$. We make use of the identity derived in Sec. B.1, i.e.,

$$\text{Tr} \left[\frac{\partial \mathcal{M}}{\partial (A_\mu)_{ji}} \mathcal{M}^{-1} \right] = \langle \eta^* \frac{\partial \mathcal{M}}{\partial (A_\mu)_{ji}} \underbrace{\mathcal{M}^{-1} \eta}_{\zeta} \rangle, \quad (\text{B.7})$$

where η is an $16(N^2 - 1)$ -dimensional white-noise vector obeying $\langle \eta_a^* \eta_b \rangle = \delta_{ab}$. The quantity ζ in the above equation can be estimated by solving the linear equation

$$\mathcal{M} \zeta = \eta, \quad (\text{B.8})$$

using the bi-conjugate gradient stabilized method (BiCGSTAB). Here, we note that finding solution to the matrix equation is always much faster than the computation of inverse. To summarise, we have reduced the explicit inversion of the large matrix $\mathcal{M}$ to a bunch of matrix multiplications and matrix equation solving.

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